Validated Synthetic Patient Generation for Small Longitudinal Cohorts: Coagulation Dynamics Across Pregnancy

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Abstract

Small longitudinal clinical cohorts, common in maternal health, rare diseases, and early-phase trials, limit computational modeling: too few patients to train reliable models, yet too costly and slow to expand through additional enrollment. We present multiplicity-weighted Stochastic Attention (SA), a generative framework based on modern Hopfield network theory that addresses this gap. SA embeds real patient profiles as memory patterns in a continuous energy landscape whose induced distribution places one component on each stored patient, and generates novel synthetic patients by sampling it while preserving the geometry of the original cohort. We generated the reported cohort with Langevin dynamics. The same distribution is also available in closed form, and an exact analytic sampler reproduced the same fidelity and privacy results. Per-pattern multiplicity weights enable targeted amplification of rare clinical subgroups at inference time without retraining. We applied SA to a longitudinal coagulation dataset from 23 pregnant patients spanning 72 biochemical features across 3 visits (pre-pregnancy baseline, first trimester, and third trimester), including rare subgroups such as polycystic ovary syndrome and preeclampsia. Across multiple validation tests, including an ordinary differential equation model of the coagulation cascade, synthetic patients generated by SA were not distinguishable from their real counterparts by the statistical, structural, and mechanistic measures applied at this sample size. A downstream utility test further showed that a mechanistic model calibrated entirely on synthetic patients predicted held-out real patient outcomes as well as one calibrated on real data. These results demonstrate, as a proof-of-concept, that SA can produce synthetic cohorts useful for the modeling use-cases tested here (mechanistic calibration and hypothesis generation) from very small longitudinal datasets, enabling data-augmented modeling in small-cohort settings.

Keywords: synthetic data generation, stochastic attention, modern Hopfield networks, multiplicity weighting, coagulation, pregnancy, longitudinal data, mechanistic validation

1 Introduction

In 2021, the United States maternal mortality rate reached 32.9 deaths per 100,000 live births, the highest level since 1965[1,2]. Between 2018 and 2022, Non-Hispanic Black women died at 2.8 times the rate of White women, and disorders related to pregnancy, including hemorrhage and venous complications, were the leading underlying cause of death, accounting for 17.4% of 6,283 pregnancy-related deaths[3]. These figures highlight the role of the coagulation system in maternal outcomes. Blood coagulation is a complex enzymatic cascade in which a small tissue factor (TF) stimulus triggers a series of serine protease activation reactions that ultimately produce thrombin, the central effector of hemostasis[4,5]. Thrombin cleaves fibrinogen into fibrin, activates platelets, and amplifies its own production through positive feedback via Factors V, VIII, and XI, while simultaneously initiating its own shutdown through the thrombomodulin–protein C anticoagulant pathway[6,7]. Pregnancy alters this hemostatic balance, shifting toward a hypercoagulable state that prepares for delivery[8–10]. Procoagulant factors including fibrinogen, Factor VIII, and von Willebrand factor increase substantially across gestation, while natural anticoagulants such as antithrombin and protein S decline. Two conditions further shift this balance in ways that are not fully understood. Polycystic ovary syndrome (PCOS), a hormonal and metabolic disorder affecting approximately 6.6% of reproductive-age women[11], has been associated with impaired fibrinolysis and a prothrombotic tendency[12]. Preeclampsia (PE), a hypertensive disorder that complicates 2–8% of pregnancies worldwide[13], is characterized by endothelial dysfunction, platelet activation, and altered thrombin generation[14]. Mathematical models of the coagulation cascade have provided mechanistic insight into patient-specific thrombin generation[15,16]. These range from the original Hockin–Mann stoichiometric framework[6] through extensions incorporating the thrombomodulin–protein C pathway[7] to the Brummel-Ziedins 2012 (BZ2012) model with detailed activated protein C (APC)-mediated Factor Va inactivation kinetics[17]. However, applying these models to pregnancy cohorts requires longitudinal datasets capturing coagulation factors, inhibitors, and thrombin generation assay (TGA) measurements across gestation, data that is rare and expensive to collect.

The core barrier is sample size. Longitudinal pregnancy coagulation studies typically enroll tens of patients with complete follow-up across multiple trimesters, yielding datasets where the number of features p (coagulation factors, TGA parameters, viscoelastic measurements) exceeds the number of patients n. Rare complications compound this problem. Small longitudinal cohorts inevitably contain only a handful of patients with any given condition, far too few for condition-specific statistical analysis. This $n < p$ regime limits conventional approaches because covariance matrices are rank-deficient, cross-validation is unreliable, and overfitting is nearly inevitable[18]. Synthetic data generation addresses this by augmenting small real datasets with statistically faithful synthetic records, enabling downstream analyses (mechanistic modeling, clinical comparisons, hypothesis generation for rare complications) that would otherwise be infeasible[19,20]. However, existing methods face limitations in the small-cohort longitudinal setting. Sampling from a fitted multivariate normal (MVN) distribution is singular when $n < p$ and must be regularized[21], introducing bias that distorts the joint distribution. MVN also assumes Gaussian marginals and linear correlations, and provides no mechanism for conditional generation: it cannot amplify a specific subpopulation while preserving its signatures. More expressive models such as generative adversarial networks (GANs) and variational autoencoders (VAEs)[22–24], including recent longitudinal extensions[20], require substantially larger training sets and are prone to mode collapse at small n[25] (we confirm this empirically for Conditional Tabular GAN (CTGAN) at n=23 in this work). A generative method is needed that can operate directly on the geometry of a small dataset without estimating a full covariance model.

Stochastic Attention (SA) is such a framework. Rather than fitting a full covariance model or training a neural generator, SA treats the stored patient profiles themselves as the model. Building on modern Hopfield network theory[26], each profile becomes a memory pattern in a continuous energy landscape, and the attention mechanism makes that landscape a distribution centered on the stored patterns. Sampling it yields new profiles rather than copies, because the memory-centered components overlap in the reduced space. Because sampling occurs in a principal component analysis (PCA)-reduced linear subspace anchored on the observed cohort, SA never forms a full $p \times p$ covariance matrix and so avoids both the rank deficiency that limits MVN and the mode collapse that limits GAN/VAE approaches when $n < p$. We recently introduced a multiplicity-weighted extension of the Hopfield energy that attaches a per-pattern weight to each stored profile[27], which provides exactly the lever that MVN lacks: continuous interpolation between unconditioned generation and targeted amplification of a rare subpopulation at inference time, without retraining. What remains open, and what this study addresses, is whether SA can faithfully generate *longitudinal* trajectories from a small cohort. In earlier proceedings work, we paired our mechanistic coagulation pipeline with per-visit MVN sampling to produce synthetic populations[28]; that approach captured single-visit marginals but, by fitting each visit independently, generated patients whose trajectories across gestation were statistically incoherent. A generator that respects the joint structure across visits is what a small longitudinal pregnancy cohort actually demands.

In this study, we used multiplicity-weighted SA to generate complete longitudinal patient profiles from a small pregnancy coagulation cohort (K=23 patients, 72 features per visit, 3 visits) that includes rare clinical subgroups (PCOS, n=3; Developed PE, n=5). Our pipeline concatenates each patient's multi-visit profile into a single vector, applies PCA for dimensionality reduction, and generates new patients via multiplicity-weighted Langevin dynamics on the Hopfield energy landscape, with a direction-magnitude decomposition that preserves the natural variance structure of continuous clinical measurements. We then asked whether the synthetic patients reproduced marginal feature distributions, the cross-visit covariance structure, condition-specific signatures of rare clinical subgroups, and the biological plausibility expected by an independent ordinary differential equation (ODE) model of the coagulation cascade[17]. As a downstream-utility test, we further asked whether a mechanistic model calibrated entirely on synthetic patients could predict held-out real patient outcomes. The results, presented in the next section, show that SA can, as a proof-of-concept, recover both the statistical structure and the mechanistic plausibility of the real cohort at this sample size. If this generalizes, the implication is that the bottleneck for studying rare obstetric and pediatric conditions may be shifting from cohort size toward cohort fidelity: a few dozen carefully phenotyped longitudinal patients, augmented through SA, may be enough to support mechanistic and statistical analyses that have traditionally relied on much larger cohorts.

2 Results

We generated N=100 synthetic longitudinal patient profiles using SA and compared them against the K=23 real patients across four validation levels. This sample size provided approximately 4× amplification while remaining in the regime where SA produces diverse, non-degenerate samples. As a baseline, we also generated N=100 patients from a regularized multivariate normal (MVN) distribution fitted to the same data.

2.1 Marginal Plausibility

We first assessed whether SA-generated patients reproduced per-feature summary statistics across all three visits. The pooled median mean relative error (MRE) across all 216 feature–visit entries was 1.2% (95% bootstrap CI: 1.0–1.6%), with 193 of 216 entries (89%) below 5% (Table S5), indicating that SA captured the central tendencies of the real population. Per-visit MREs for representative coagulation features are summarized in Table 2, where Factor II and Factor VIII fell below 2%, antithrombin below 3%, and fibrinogen below 1% across all three visits. We verified that synthetic patients were not memorized copies of real patients using two complementary metrics summarized in Table S9. The mean novelty score ($1 - \max_k \cos(\hat{\boldsymbol{\xi}}, \hat{\mathbf{m}}_k)$) was 0.50 at $\beta = \beta^*$, indicating that generated samples were on average 50% away from their nearest stored pattern in angular distance. The median nearest-neighbor distance from each synthetic patient to its closest real patient was 6.14 standardized units, compared with 6.87 for the median real-to-real nearest-neighbor distance, a ratio of 0.89 indicating that synthetic patients were approximately as far from real patients as real patients were from each other.

We next examined whether known physiological relationships and longitudinal trajectories were preserved in the synthetic cohort. Pairwise scatter plots of coagulation factor levels against thrombin generation parameters showed that SA-generated patients occupied the same joint regions as real patients (Fig. 1). The expected inverse relationship between antithrombin and TF-initiated peak thrombin was preserved, as were the positive associations between Factor VIII and peak, prothrombin and endogenous thrombin potential (ETP), and the inverse relationship between antithrombin and ETP. Per-visit means and standard deviations for six key coagulation features confirmed that synthetic patients tracked the real longitudinal trajectories closely, with overlapping confidence bands at all three visits (Fig. 2). Fibrinogen, Factor VIII, and von Willebrand factor showed the expected monotonic increases from baseline through the third trimester in both cohorts, while Factor II increased modestly and antithrombin declined then partially recovered, consistent with the known hypercoagulable shift of pregnancy. Together, these results indicated that SA captured not only univariate distributions and pairwise dependencies but also the directionality and magnitude of longitudinal change. The 72 features also included viscoelastic (ROTEM) and fibrinolytic parameters, which were generated as part of the concatenated vector and showed comparable MREs to the coagulation factors (Table S5), but are not individually highlighted here because the mechanistic validation uses the BZ2012 thrombin generation model, which does not cover fibrinolysis or clot viscoelasticity. These features still load onto the retained principal components, so they shape the energy landscape from which every synthetic patient is drawn. Their statistical fidelity matters even though we do not check them mechanistically. A fibrinolysis model under development will close that gap.

We also compared SA against two widely used deep generative methods for tabular data, Conditional Tabular GAN (CTGAN) and Tabular Variational Autoencoder (TVAE)[24] (Table S6). CTGAN failed completely, producing median MREs of ≈19%, an order of magnitude worse than SA, across all epoch counts tested (300–3,000), consistent with known GAN mode collapse on small datasets. TVAE achieved comparable marginal fidelity at high epoch counts (median MRE 1.8% at 3,000 epochs), but operated on per-visit rows (n=90) rather than concatenated longitudinal profiles (n=23), meaning it could not capture cross-visit covariance structure and had no mechanism for conditional subgroup generation.

We next asked how much of this fidelity comes from the shared principal-component subspace and how much from the stochastic-attention dynamics. We compared SA against four generators fit in the identical d=18 PCA subspace, a Gaussian mixture, a kernel-density estimator, k-nearest-neighbor interpolation, and a Gaussian copula (Supplementary Table S13). The subspace proved necessary but not sufficient. Two of the four baselines failed despite sharing it. k-nearest-neighbor interpolation collapsed novelty to a median of 0.05 and essentially returned the training patients, and the kernel-density estimator degraded both marginal fidelity (1.9% median relative error) and mechanistic plausibility (0.82 cloud overlap). The mixture and copula baselines cleared both bars. They matched SA on marginal error (1.1% and 1.3% against SA's 1.2%) and on mechanistic overlap

(0.92 and 0.93 against 0.90), and only SA and the copula put every synthetic patient above the novelty threshold. The subspace therefore supplies much of the marginal and mechanistic fidelity, and any generator that samples it sensibly inherits that fidelity. We take this as the substantive answer to whether the representation or the dynamics carries the result.

The tie did not extend to the remaining axes. SA showed the largest empirical cross-visit correlation error of the five methods (0.64 against 0.42 for the mixture and copula). That quantity measures reproduction of the correlation matrix of the same 23 patients every generator was fit to. At K=23 that matrix is a noisy estimate of the population structure, so reproducing it does not test generalization. The simulation benchmark does, because there the population covariance is known by construction, and SA recovered it more accurately than the multivariate normal baseline across the $n < p$ regime (Fig. 4A). SA also has two properties the subspace baselines lack. It fixes its operating point from the attention-entropy inflection instead of tuning a bandwidth or component count on 23 patients, and it admits the inference-time multiplicity steering used below for conditional subgroup generation.

2.2 Cross-Visit Covariance Structure

Having established that SA reproduces marginal distributions and pairwise relationships, we next asked whether it preserves the higher-order joint structure across visits. We compared the full 216×216 cross-visit correlation matrices for real, SA-generated, and MVN-generated populations. The real data exhibited a characteristic block structure with strong within-visit correlations along the diagonal and structured off-diagonal blocks reflecting cross-visit dependencies; for example, a patient's Visit 1 Factor X level predicting their Visit 3 Factor X level. We found that SA preserved this block structure, including the off-diagonal cross-visit correlations (Fig. 3). The SA−Real residual matrix showed small, unstructured deviations distributed throughout, whereas the MVN−Real residual was notably smoother, reflecting the regularization-induced shrinkage of off-diagonal correlations toward zero. This difference was most apparent in the off-diagonal blocks, where MVN systematically underestimated cross-visit dependencies that SA retained. We note that Pearson correlation captures linear associations; Spearman rank correlations, used in the mechanistic validation, would additionally capture monotonic nonlinear dependencies.

We examined the eigenvalue spectrum of the sample covariance matrix to understand the structural origin of this difference (Fig. S2). The real data had rank 22, reflecting the K=23 patient constraint. SA and MVN make different trade-offs in handling this rank deficiency: SA truncates via PCA (zeroing variance beyond component 18), while MVN regularizes via Ledoit–Wolf shrinkage (providing a full-rank estimate by inflating eigenvalues in the 194 null dimensions). The consequence of MVN's approach is that it introduces variance in dimensions where the data contain no signal, which manifested as increased dispersion in PCA projections. PCA projections by visit confirmed this dispersion gap (Supplementary Fig. S8). SA-generated patients occupied the same region of PCA space as real patients across all three visits, while MVN-generated patients showed substantially greater scatter, particularly at Visits 2 and 3.

2.3 Robustness Across Data Regimes

We next tested whether SA recovers population structure or merely reproduces the training cohort. This requires a synthetic ground truth, where the true covariance is known and does not depend on any particular draw. We swept a low-rank Gaussian generative model across sample size n, dimensionality p, and latent rank. SA recovered the population covariance more accurately than a multivariate Gaussian throughout the rank-deficient $n < p$ regime (Fig. 4A). At the real cohort's shape (n=23, p=216) the Gaussian's relative covariance error was about 1.5 times SA's, and SA was more accurate in 35 of the 39 Gaussian $n < p$ cells. The target covariance is fixed in advance here, so this is generalization and not memorization. The empirical cross-visit comparison could not draw that distinction. The advantage narrowed as n approached and exceeded p, where the Gaussian is well specified, so the benefit belongs to the small-sample regime this cohort occupies.

We also subsampled the real cohort to as few as eight patients and regenerated. Performance degraded gracefully. Recovery of the full-cohort cross-visit correlation structure worsened smoothly as n fell (Fig. 4B). The pooled marginal error rose from about 2% at n=20 to about 5% at n=8, and mechanistic plausibility stayed near 0.90 at every subset size. We then tested a deliberately non-linear ground truth, an S-curve manifold. As curvature increased, SA's linear principal-component subspace placed generated points off the true manifold, and the off-manifold residual reached twice its linear-baseline value at a curvature of 0.25 to 0.5. SA still stayed closer to the manifold than the multivariate Gaussian, and the Gaussian recovered the gross covariance better. Neither method wins on both counts. The linear embedding is a bounded choice matched to the cohort size.

2.4 Interpolation and Extrapolation

We next asked whether the sampler stays within the real cohort or reaches beyond it. We tested convex-hull membership of the synthetic patients against the 23 real patients in the d=18 principal-component space (Supplementary Table S14). Every synthetic patient fell outside the hull. So did every real patient with respect to the other 22. We then matched the comparison for severity by removing each point's nearest vertex on both sides, so that real and synthetic patients faced equally depleted hulls. Matched this way, the distance to the hull was not distinguishable between them. The hull criterion did not separate synthetic patients from real ones at K=23.

The reason is geometric. With 23 points in 18 dimensions the hull is nearly a simplex. It reaches the patients' typical radius only along the 23 vertex directions and extends a small fraction of that distance in a generic direction, so almost any point that does not sit on a stored pattern lies outside it. The sampler compounds this. It normalizes the stored patterns to unit length and returns each sample to the unit sphere, and a unit vector inside the convex hull of unit-norm patterns must coincide with one of those patterns. Hull membership is unavailable to this generator by construction, not merely uncommon. The hull criterion reports its own resolution at K=23, and we do not read it as evidence of runaway extrapolation. We assessed the generated points against the plausibility criteria instead. Most satisfied the biological constraints, and per-feature mechanistic ratios stayed inside the real cohort's percentile band for most patients. Fewer than half met the strict criterion, which requires every feature and visit to fall inside the band at once.

2.5 Re-identification Risk

The energy landscape places a basin at each stored patient, so we asked how much the generated cohort reveals about who was in it. For every synthetic patient we measured the distance to the closest real record and compared it against the leave-one-out distance between real patients. We also ran a nearest-neighbor membership-inference attack, which retrains the generator on a subset of the cohort and scores each real record by its proximity to the resulting synthetic set (Supplementary Table S15). Synthetic patients sat slightly closer to the real cohort than real patients sat to one another. The attack separated training members from held-out patients with a mean AUC of 0.97 across 40 random splits, and reached perfect separation on roughly a fifth of them. At the published operating point the generator reproduces its training cohort closely enough for membership to be recovered.

Decomposing the attack shows what it does and does not recover. Held-out patients sat at the generic distance between two unrelated patients from the synthetic cohort, matching the leave-one-out distance among the real patients themselves. Training members sat closer. The attack therefore recovers which patients were stored in the memory. It recovers nothing about patients the generator never saw, and the separation comes from the same small-cohort geometry that left the hull criterion uninformative. The mechanism is visible in the form of the target. The distribution the energy induces is a mixture with one component centered on each stored patient (Eq. (3)). Every draw descends from a single stored patient and carries proximity to that patient into the decoded profile. Nothing in the construction averages membership across several patients. The attack also assumes an adversary who already holds a patient's record and asks whether it was in the training cohort. It bounds membership disclosure, and does not show that patient-level values can be reconstructed from the synthetic cohort alone. We also tested whether the leak comes from the target distribution or from the sampler. Metropolis-adjusted sampling left it unchanged, and so did a wide sweep of the inverse temperature. Exact ancestral draws from Eq. (3) use no chain at all, and they reproduced the signal at an AUC of 0.98 (Supplementary Table S16, Fig. S9). At K=23 the signal belongs to the target distribution, not to the sampler.

2.6 Conditional Generation of Rare Subgroups

The previous analyses used unconditioned SA, generating from the full cohort. A key clinical application, however, is generating patients from specific subpopulations that are too small to study independently. We therefore tested whether SA could amplify rare clinical subpopulations while preserving their condition-specific signatures. We defined three overlapping subgroups by pregnancy outcome and comorbidity: Uncomplicated (n=18), PCOS (n=3, cross-cutting; 1 patient also in the PE group), and Developed PE (n=5). The PCOS and Developed PE groups were too small for any independent statistical analysis. We used SA's multiplicity-weighted sampling to generate condition-specific cohorts of 100 patients each by upweighting attention on the relevant stored patterns.

We compared condition-specific feature means between real and SA-generated patients across eight coagulation features that exhibited between-condition variation (Fig. 5). SA-generated cohorts preserved the condition-specific patterns. PCOS patients showed elevated Factor VIII and vWF relative to Healthy patients in both real and synthetic data, PE patients showed elevated α_2-AP and Factor IX, and the between-condition rank ordering was maintained across all eight features. To quantify equivalence, we performed a bootstrap Mann–Whitney test. For each feature and condition, we subsampled the synthetic cohort to match the real sample size (n_{real}), applied a Mann–Whitney U test, and repeated 1,000 times. We reported the fraction of replicates where $p > 0.05$ (i.e., real and synthetic were statistically indistinguishable). Across all 24 feature–condition pairs, 20 (83%) achieved $p > 0.05$ in $\geq$90% of replicates, with a median non-significance fraction of 98.6% (full per-pair results in Table S10). The four features that fell below 90% were high-variance features in the smallest subgroups (FIX and plasminogen in PCOS; FIX and plasminogen in Developed PE), where the real data exhibited large inter-patient variability. No multiple-testing correction was applied to the 24 simultaneous comparisons; the failing pairs had consistently low non-significance fractions (62–87%) rather than borderline values, so correction would not change the classification. PCA projections of the conditioned cohorts are provided in Figs. S3–S4. These results showed that SA's multiplicity-weighted interpolation can amplify rare subgroups in a way that preserves their distinguishing clinical features, useful for hypothesis generation and power analysis in small patient populations. MVN has no mechanism for this: fitting a separate MVN to three PCOS patients across 216 features is mathematically impossible (rank 2 covariance), and post-hoc subsetting of unconditional MVN samples does not preserve subgroup-specific structure.

The amplification is bounded by how few patterns underlie it. At the target effective fraction the participation ratio was 4.6 for the PCOS cohort and 7.7 for the Developed PE cohort (Table S2). Each cohort of 100 generated patients is a recombination of three and five real individuals. Those cohorts carry no more information about the conditions than the underlying patients do. Any downstream analysis that treats them as 100 independent observations will understate uncertainty. What conditioning provides is a denser sample of the distribution implied by those few patients. That supports hypothesis generation and power calculation. It is not additional evidence about the conditions themselves.

2.7 Mechanistic Consistency

The preceding validations assessed statistical properties of the synthetic data. We next tested whether SA-generated patients produce biologically plausible outputs when their coagulation factor levels are fed through a mechanistic model that knows nothing about the SA generation process, though it was calibrated on the same cohort. We used the Hockin–Mann BZ2012 model[6,17], a system of 58 ordinary differential equations with 64 rate constants that simulates thrombin generation from patient-specific coagulation factor inputs. We calibrated 5 of the 64 rate constants (prothrombinase, intrinsic and extrinsic tenase, protein C activation, and meizothrombin conversion; Table S4) on Visit 1 real patients at the population level and held the remaining 59 at published literature values. We then ran the calibrated model on all 23 real patients and 100 synthetic patients under both TF-only and TF+TM conditions (738 total simulations, 0 failures). For each patient, we computed five ODE-predicted thrombin generation features (lagtime, peak, time-to-peak, maximum rate, and ETP) and compared them against the corresponding values from the patient record. In these comparisons (Fig. 6, top row), the horizontal axis is the TGA value from the patient record, which exists for both real and synthetic patients, while the vertical axis is the value computed by the BZ2012 model from that patient's factor inputs. Systematic deviations from the $y{=}x$ line reflect ODE model bias (e.g., lagtime is underpredicted at $\approx 0.77\times$), not a failure of the synthetic data; the key observation is that real and synthetic patients share the same pattern of model bias, occupying the same cloud with the same systematic offset.

We formalized this observation by computing the ODE-predicted/dataset ratio for each patient and comparing the resulting distributions between real and synthetic populations (Fig. 6, bottom row). These ratio distributions capture how the ODE model processes each patient's factor levels. If SA had generated patients with biologically implausible factor combinations, the model would process them differently, producing a shifted or broadened ratio distribution. Instead, the distributions overlapped substantially, with cloud overlap (fraction of synthetic ratios within the 5th–95th percentile of real ratios) ranging from 0.83 for ETP to 0.92 for T.Peak under TF-only conditions, and two-sample Kolmogorov–Smirnov (KS) tests did not detect a difference between the real and synthetic ratio distributions at this sample size across all five TGA features (D = 0.081–0.123, all $p > 0.30$; full diagnostics in Table S11). Under TF+TM conditions, the model systematically overcorrected the protein C anticoagulant pathway, producing peak and maximum rate predictions approximately $0.5\times$ measured values for both real and synthetic patients (Fig. S5, same layout as the main-text comparison), yet this shared systematic bias was itself informative. Cloud overlap remained high (0.88–0.92 across the five features), confirming that the model processed real and synthetic patients identically even under conditions where the calibration was imperfect.

The calibration also generalized across pregnancy timepoints. Although fitted only on Visit 1 real patients, the per-visit median predicted-to-measured ratios under TF-only conditions remained close to 1.0 at Visits 2 and 3 (Fig. S6), confirming that the calibrated rate constants captured a stable population-level property of the coagulation system rather than overfitting to the training visit. Spearman rank correlations between predicted and measured values were moderate for ETP ($\rho \approx 0.6$–0.8 for real, $\rho \approx 0.6$–0.65 for synthetic; Fig. S7) and weak for other features, consistent with a 5-parameter population-level calibration that was not designed to resolve inter-patient variability. The key finding was not patient-level prediction but population-level plausibility. A mechanistic model of the coagulation cascade placed SA-generated patients within the same biologically plausible envelope as real patients, and could not distinguish them under either experimental condition. That model is structurally independent of the generator, though it was calibrated on the same cohort.

2.8 Downstream Utility: Mechanistic Model Calibration

The preceding results established that SA-generated patients are statistically and mechanistically plausible. To test whether this plausibility translates to practical utility, we calibrated a mechanistic model entirely on synthetic patients and evaluated whether it could predict real patient outcomes. We calibrated the BZ2012 model (TF-only condition, 5 rate constants) on synthetic V1 patients (N=100) using the same bounded Nelder–Mead optimization applied to the real V1 calibration (K=23), with 11 random restarts to mitigate local-minimum sensitivity (Table S4 for parameter details). We then evaluated both calibrations on held-out real V2 and V3 patients that neither model had seen during training. The synth-calibrated model achieved comparable or slightly better performance across all five TGA features, with per-feature median relative errors 2–10% lower than the real-calibrated model (overall ratio 0.94×; Fig. 7; per-feature breakdown in Table S12). This improvement likely reflects the smoother loss landscape afforded by N=100 synthetic training patients compared with K=23 real ones, reducing overfitting during optimization. The two calibrations found different parameter values but produced highly correlated predictions across all five TGA features, confirming that the synthetic data captured the same underlying coagulation structure. We note that the synthetic training data is one step removed from the real data (real → SA → synthetic → calibration), so the synth-calibrated model is not fully independent of the real data; rather, it demonstrates that SA's representation preserves sufficient biological structure for a mechanistic model to learn generalizable parameters.

The direction of that comparison should not be overread. Every piece of information in the synthetic training set derives from the same 23 patients. The modest advantage is a variance-reduction effect, not new biological content. Fitting five rate constants against 100 samples drawn from a distribution estimated on 23 patients smooths the objective and dilutes the influence of any single patient. Any generator producing a denser sample of the same distribution would do the same. The result establishes that the synthetic cohort is a usable substitute for calibration, not a better one.

Read alongside the conditioned cohorts, this bounds what the method offers in practice. It enables analyses that the original sample size forecloses. We calibrated a mechanistic model without spending the real cohort on training, and we compared subgroups holding three and five patients. It also points to factor-level differences a larger study could test. It does not substitute for additional patients. None of these results speak to clinical deployment, which would require prospective validation against real clinical decisions.

3 Discussion

This study tested whether a geometry-preserving generative method could produce synthetic patients from a very small longitudinal cohort that were not merely statistically similar to real patients but consistent with the underlying coagulation biology. Our results provided evidence at each validation level that SA-generated patients met this standard. Individual feature distributions were reproduced with median MRE of 1.2%, which MVN also achieved by construction. SA additionally preserved the joint cross-visit covariance structure that MVN's rank-deficient estimate could not represent, a difference made visible in the eigenvalue spectrum where MVN introduced spurious variance in 194 null dimensions while SA respected the low-rank geometry of the data. SA could amplify clinically meaningful rare subgroups, generating 100 synthetic PCOS patients from only 3 real ones, while preserving subgroup-specific signatures that MVN cannot capture. And an independent ODE model of the coagulation cascade, calibrated exclusively on real patients, confirmed that the factor-to-thrombin-generation mapping in synthetic patients was biologically plausible, with 83–92% cloud overlap and Kolmogorov–Smirnov tests unable to distinguish the two populations ($p > 0.35$ for all five TGA features).

The ability to generate validated synthetic cohorts from very small longitudinal datasets has practical implications for maternal health research. Rare pregnancy complications such as PCOS and preeclampsia are difficult to study because assembling large, well-characterized longitudinal cohorts requires years of recruitment across multiple clinical sites. Our results suggest that SA can enable hypothesis generation, power analyses, and computational modeling studies for these understudied populations by generating synthetic cohorts that preserve both the statistical and mechanistic properties of the real data. The conditional generation capability is particularly relevant. Rather than requiring researchers to collect additional rare patients, SA can amplify existing small subgroups into cohorts large enough for meaningful analysis.

Understanding why SA succeeds where parametric methods fail requires examining the geometric structure of the problem. SA generates samples by interpolating between stored patient profiles via the Hopfield energy landscape operating in a PCA-reduced linear subspace, rather than fitting a parametric distribution that must regularize or truncate high-dimensional structure. In the $n < p$ regime that characterizes most small clinical studies, any parametric distribution faces a fundamental identifiability problem. There are more parameters to estimate than observations to estimate them from. SA sidesteps this by operating in a PCA-reduced space where the memory-to-dimension ratio is favorable ($K/d_{\mathrm{PCA}} \approx 1.28$) and by preserving the full directional structure of the data through the attention mechanism. A natural concern is whether the β^* selection procedure, originally developed for protein sequence generation[29], transfers to clinical coagulation data. The entropy inflection method identifies the phase transition in the Hopfield energy landscape, which is a geometric property of K patterns in d dimensions, not a property of what the features represent; for our dataset the empirical inflection ($\beta^* \approx 2.94$) was close to the theoretical prediction ($\sqrt{d} \approx 4.24$), and a sensitivity analysis confirmed that generation quality degraded gracefully across $\beta/\beta^* \in [0.1, 3.0]$ (Table S1). Similarly, varying the PCA variance retention threshold from 85% to 99% ($d_{\mathrm{PCA}} = 13$ to 21) had minimal impact on generation quality. Median MRE remained between 1.2% and 1.8% across all thresholds (Table S7), confirming that the 95% threshold was not a critical design choice. The multiplicity weighting for conditional generation introduced an additional consideration. Achieving 80% attention on the PCOS subgroup required $\rho \approx 26.7$, reducing the effective pattern count to $K_{\mathrm{eff}} \approx 4.6$ (multiplicity parameters for all three subgroups in Table S2), which explains the larger MREs observed for PCOS features because the energy landscape was effectively shaped by fewer than 5 independent patterns, pulling means toward the population average. The direction-magnitude decomposition compounds this limitation. For PCOS, magnitudes are drawn from only 3 empirical norms, providing minimal diversity in the scale of generated samples. More broadly, the SA framework and its multiplicity-weighted conditioning have been demonstrated in companion studies from this group on discrete protein sequence generation from small family alignments[29], on steering generation toward functional subsets such as binding peptides[27], and on the theoretical foundations connecting modern Hopfield networks to attention-based generation[30]. Across these domains, the critical temperature prediction ($\beta^* \approx \sqrt{d}$) and the multiplicity-weighted conditioning mechanism transfer without modification, suggesting that the geometric properties exploited here are not specific to clinical coagulation data but are general features of the Hopfield energy landscape operating on small pattern sets.

An alternative precedent for validated longitudinal synthetic generation is the VAMBN-MT framework of Kühnel et al.[20], which extends a Variational Autoencoder Modular Bayesian Network with an LSTM time-encoder to better capture cross-visit dependencies. They evaluated VAMBN-MT on the DONALD nutritional cohort and on the Alzheimer's Disease Neuroimaging Initiative (ADNI) cohort. Their study is instructive because it offers a regime contrast to ours. With N=1,312 DONALD participants and 33 longitudinal variables, their setting sits in the data-rich regime for which VAMBN-MT was designed, where deep parametric models can be trained directly on the cohort. The published configuration combines expert-curated module assignments, Bayesian-network edge constraints, an LSTM-augmented HI-VAE per module, and 10,000 synthetic samples drawn from the trained model to reproduce a published age-trend regression. In their evaluation, the best variant reported a cross-visit correlation matrix relative Frobenius error of roughly 0.70, and downstream time trends weaker than the strongest signals in the real data were not reproduced at the sample sizes they tested, both honest indicators of how hard this generative task is even in the data-rich regime. On the other hand, our setting inverts both the number of patients and the number of features. With K=23 patients across 216 features we operate two orders of magnitude smaller and squarely in the $n < p$ regime, well below the cohort sizes for which VAMBN-MT was developed. The non-parametric design of SA addresses this regime in three ways that contrast with VAMBN-MT. First, PCA-derived linear geometry replaces the modular HI-VAE architecture as the mechanism for capturing cross-visit dependencies. Second, a generic mechanistic-model validation pattern, transferable to any domain with a calibrated ODE model, replaces the domain-specific dependency rules used in their evaluation, such as graduation-status monotonicity, age-time arithmetic identities, and macronutrient summation. Finally, multiplicity weighting provides an inference-time conditional-generation lever that the VAMBN-MT framework does not offer. These are complementary positions in the design space rather than a head-to-head comparison. VAMBN-MT addresses the moderate-cohort regime where deep generative models are feasible but expert-specified modular structure helps capture cross-visit dependencies, while SA addresses the small-cohort regime where parametric models cannot be fit and the geometry of the few real patients must itself carry the generative signal.

Beyond these statistical and geometric considerations, the mechanistic validation introduced an approach that applies beyond coagulation. Rather than relying solely on statistical comparisons between real and synthetic distributions, we tested whether synthetic patients produced biologically consistent outputs when processed by an independent computational model, and went further to show that a mechanistic model calibrated entirely on synthetic data predicted real patient outcomes as well as one calibrated on real data (overall ratio 0.94× on held-out V2+V3 patients, with 11 random restarts per calibration). This approach is available whenever a validated mechanistic model exists for the domain of interest (pharmacokinetic models in drug development[31], physiological models in critical care[32], tumor growth models[33], or metabolic models[34,35]), and the requirement is not that the model be a perfect predictor but that synthetic patients fall within the same model-predicted envelope as real patients. The bootstrap Mann–Whitney analysis of condition-specific features revealed that 5 of 24 (21%) feature–condition pairs were statistically distinguishable at equal sample sizes, but the mean relative errors for these features remained small (5–15%), and the detected differences likely reflected SA's tendency to compress distributional tails rather than shift means. This tail compression arises from two sources: the PCA dimensionality reduction, which cannot fully represent higher-order moments of every feature, and the direction-magnitude decomposition, which draws magnitudes from a finite empirical distribution of K=23 norms. This is a trade-off. SA sacrifices some tail fidelity in exchange for preserving the manifold geometry that parametric methods cannot capture in the $n < p$ regime.

The clinical reading of that trade-off matters, because extreme states are often the phenomena of interest. If a generated cohort under-represents the severe end of a distribution, estimates of how often a severe phenotype occurs, and of how extreme it can become, will be biased toward the center. The extrapolation analysis shows that the generator does place patients outside the convex hull of the real cohort, so it is not confined to the observed range. But producing out-of-range points is not the same as reproducing how often extreme ones occur, and the two should not be conflated. We recommend against using SA cohorts to estimate extreme-quantile risk without real anchoring data. The mechanistic checks reported here are less sensitive to this limitation. They compare distributions within a fifth to ninety-fifth percentile band, so they test the central mapping from factor levels to thrombin generation and not the tails.

Several of the findings above are one observation, not a list. With 23 patients in a 216-dimensional measurement space, the cohort is sparse enough that familiar diagnostics stop discriminating. Convex-hull membership rejects every synthetic patient, but it rejects every real patient with respect to the others as well. It reports the resolution of the criterion, not a property of the generator. The stored patterns sit far enough apart that their basins do not overlap. That is why a membership-inference attack succeeds, and why conditioning on three PCOS patients leaves an effective pattern count below five. The advantage of the synthetic cohort in downstream calibration is a variance-reduction effect, obtained by sampling more densely from a distribution estimated on those same patients. And the shared principal-component subspace supplies much of the marginal and mechanistic fidelity, which is why several baselines fit in that subspace match SA on those axes. Each of these follows from the sample size. None is a separate defect of the method.

This study had several limitations. Our study used a single longitudinal dataset of K=23 patients; further evaluation on independent datasets and across different clinical domains is needed to establish generalizability. We addressed this in two partial ways, short of a second clinical cohort. The simulation benchmark supplies a ground truth that no single dataset can. There the population covariance is known in advance, so generalization can be separated from reproduction of the sample. Separately, the same energy landscape, operating-point rule, and multiplicity conditioning have been applied to discrete protein sequence generation from small family alignments and to steering generation toward functional subsets. The mechanism exploited here is geometric, not specific to coagulation. Neither argument substitutes for replication in an independent clinical cohort. The PCA dimensionality reduction assumed linear structure in the feature space; nonlinear manifold learning methods may better capture complex clinical relationships, though at the cost of reduced interpretability. The mechanistic validation under TF+TM conditions revealed systematic ODE model bias, though the observation that real and synthetic patients shared the same bias pattern was itself informative. The BZ2012 calibration required substantial departures from published rate constants, notably a 50-fold reduction in intrinsic tenase k_{cat} and a 15-fold increase in extrinsic tenase k_{cat} (Table S4). These deviations likely reflect differences between the *in vitro* conditions under which the literature rate constants were measured (purified component systems) and the plasma-based TGA used in this study, where phospholipid surfaces, factor concentrations, and inhibitor profiles differ from the model assumptions[6,17]. The purpose of the mechanistic validation was not to obtain biochemically accurate rate constants but to test whether the ODE model processed real and synthetic patients equivalently, a comparison that is invariant to the absolute parameter values. The ODE was calibrated on this same cohort. The mechanistic agreement demonstrates consistency with a model fitted to the same biological system. It is not validation against a fully independent ground truth. Its evidential value rests on the model being blind to whether a profile is real or synthetic, and on its applying one fixed mapping to both. The mechanistic validation was limited to thrombin generation (BZ2012) and did not cover the fibrinolytic or viscoelastic features included in the dataset; extending this validation to a mechanistic model of fibrinolysis, currently under development, would provide additional evidence for these feature categories. Finally, we did not validate synthetic patients against clinical outcomes; our validation was restricted to statistical and mechanistic plausibility, and prospective validation against real clinical decision-making remains an important future direction.

The concatenation that makes cross-visit structure available also constrains the data the method accepts. Stacking three visits into a single vector assumes that every patient has every visit, and that visits are aligned across patients. This cohort satisfies both by design. Most observational data does not. Irregular timing would place physiologically different states in the same coordinate, so the principal components would mix visit structure with between-patient variation. Missing assays and missing visits have no natural representation in a fixed-length vector. They would require imputation before the embedding, which at $K{=}23$ would itself carry large uncertainty. Variable-length records fall outside the formulation altogether. They would need padding to a common length, or a sequence-aware extension in which attention operates over visits as well as over patients. These are scope boundaries of the present formulation. The energy landscape itself is indifferent to what the coordinates mean.

Cost and behavior at larger cohort sizes follow from how the target is sampled. In the Langevin implementation, each step evaluates the attention weights over the stored patterns and forms their weighted combination. The work per step is the pattern count times the retained dimension, and one sample costs that product times the number of steps. The closed form removes the sequential factor entirely for future use. Drawing a component and adding isotropic noise costs one selection and one Gaussian draw per sample, whatever the step count. The remaining work is the principal-component decomposition and the decoding, and both are shared across the cohort. Either way the method is linear in the number of patients and in the reduced dimension. At high feature counts the practical bound is the principal-component decomposition, not the sampling. Statistically the picture runs the other way. The simulation benchmark shows that SA's advantage over a fitted Gaussian is largest when the sample is smallest, and that it narrows as the sample size approaches the dimension and the Gaussian becomes well specified. The method is therefore positioned for the small-cohort regime, not as a general-purpose generator. We expect the memorization documented above to ease as the cohort grows, because basins that currently stand apart would begin to overlap and generation would interpolate across several patients instead of staying near one. This cohort admits only subsampling downward, so we did not test that expectation.

The linear subspace deserves the same scrutiny. Principal components capture linear structure. A manifold with strong curvature, such as a branching trajectory or a threshold effect, is represented only approximately, and that approximation error is one source of the tail compression discussed above. The simulation benchmark bounds where this begins to matter. On a deliberately curved ground truth the off-manifold residual reached twice its linear-baseline value at a curvature of 0.25 to 0.5. Beyond that point SA still stayed closer to the manifold than a fitted Gaussian, and the Gaussian recovered the gross covariance better. Neither method dominates once the target stops being linear. Nonlinear embeddings such as kernel principal components, diffusion maps, or autoencoders would represent curvature more faithfully, at three costs. They give up components expressed directly in assay space and the interpretability that follows. They give up the closed-form relation between the critical temperature and the dimension that fixes the operating point here. And they introduce fitting choices that are unstable to estimate from 23 patients. Linearity is therefore a deliberate choice matched to the cohort size, not an oversight. The subspace ablation shows which behaviors follow from the representation, since generators sharing that subspace match SA on marginal and mechanistic fidelity while differing sharply in how far they move from the stored patients.

A distinct limitation concerns privacy. The generator concentrates probability mass near each of the K=23 stored patterns, so synthetic patients lie close to individual real records. A nearest-neighbor membership-inference attack separates cohort members from held-out patients with high accuracy. We find that this behavior belongs to the target distribution at K=23. It is not an artifact of the sampler or the operating temperature. Metropolis-adjusted sampling removes the discretization bias of the unadjusted scheme, and it left the signal unchanged. Lowering the inverse temperature improved the distance to the closest real record only marginally. That distance never reached the level separating real patients from one another, and buying it cost cross-visit covariance fidelity. The closed form of the target settles the attribution. The latent distribution is a Gaussian mixture whose components are centered on the stored patterns, so it can be drawn from analytically, with no chain at all. Exact draws carried through the same decoding pipeline reproduce the membership-inference signal in full. Proximity to individual real records is therefore a property of the distribution the method defines at K=23, not of how that distribution is approximated. We do not present the method as privacy-preserving. It is not differentially private. The practical risk is bounded instead by the fully de-identified source cohort, and by intended uses that do not require releasing patient-level records, such as mechanistic calibration and hypothesis generation. The burn-in and thinning used in the sampling diagnostic were derived once on the full cohort and reused for the subset analyses. The diagnostic characterizes the operating regime, and does not certify each subset independently.

4 Conclusion

We have demonstrated that multiplicity-weighted Stochastic Attention can generate synthetic longitudinal patient data from very small clinical cohorts (K=23) that pass four levels of validation: marginal plausibility, cross-visit covariance preservation, conditional subgroup amplification, and mechanistic consistency with a coagulation ODE model calibrated on the same cohort. A downstream utility test showed that a mechanistic model calibrated entirely on synthetic patients predicted real patient outcomes as well as one calibrated on real data. The SA framework preserves the geometry of the clinical data in regimes where parametric approaches such as multivariate normal sampling fail due to rank deficiency. An analytic sampler for the same latent target reproduced the same fidelity and privacy results, and offers a faster route for future use. Combined with a multilevel validation framework, this approach provides a proof-of-concept path for generating synthetic cohorts useful for the modeling use-cases demonstrated here to support computational research in rare conditions and small longitudinal studies.

5 Methods

5.1 Population Dataset

We analyzed a longitudinal dataset of coagulation, fibrinolysis, thrombin generation assay (TGA), and rotational thromboelastometry (ROTEM) measurements from women desiring a pregnancy. The study was approved by the Institutional Review Board at the University of Vermont. Written consent was obtained from all participants prior to enrollment. Blood collection, plasma processing, and assay protocols have been described previously[36–39]. Briefly, citrated platelet-poor plasma was collected without tourniquet use after supine rest at three clinical visits: Visit 1 (pre-pregnancy baseline in the follicular phase), Visit 2 (end of first trimester), and Visit 3 (mid-third trimester). Plasma aliquots were stored at $-80°$C for subsequent analysis. Measurements of secondary analytes were conducted as follows: activity levels of coagulation factors were determined by Stago clotting assays, fibrinolysis proteome levels were measured by ELISA, and hormone levels were determined by the CLIA-certified laboratory at the University of Vermont Medical Center. TGA was performed using 5 pM relipidated tissue factor with fluorogenic substrate on a Synergy4 plate reader as described in McLean et al.[36]. Clot formation and fibrinolysis dynamics were assessed via viscoelastometry using the ROTEM Delta Instrument (Werfen). Thawed plasma was mixed with or without 4 nM tPA (with respect to plasma volume), recalcified with 15 mM calcium chloride, and incubated for 3 minutes at 37°C. The reaction was initiated by adding the plasma mixture to ROTEM cups containing tissue factor at a 17:1 ratio; the final concentration of TF in the reaction was 5 pM.

The dataset contained 153 total records across approximately 50 subjects. After removing columns with $> 30\%$ missing values and retaining only subjects with complete data across all three visits, we obtained $K = 23$ complete longitudinal profiles across $n = 72$ assay measurements per visit. The 72 assay measurements spanned five categories: (i) hormones (estradiol, progesterone), (ii) coagulation factors and inhibitors (Factors II, V, VII, VIII, IX, X, XI, XII; antithrombin (AT), protein C (PC), tissue factor pathway inhibitor (TFPI), von Willebrand factor (vWF), fibrinogen, etc.), (iii) fibrinolytic markers (plasminogen, plasminogen activator inhibitor-1 (PAI-1), plasminogen activator inhibitor-2 (PAI-2), α_2-antiplasmin (α_2-AP), thrombin-activatable fibrinolysis inhibitor (TAFI), tissue plasminogen activator (tPA), D-dimer equivalents), (iv) thrombin generation parameters under four initiator conditions (TF at 5 pM, TF+thrombomodulin (TM), No TF, No TF+anti-FXIa), and (v) ROTEM/viscoelastic parameters (clotting time (CT), maximum clot firmness (MCF), alpha angle, lysis onset time (LOT), maximum lysis (ML), lysis time (LT), area under the curve (AUC) under TF Only and TF+tPA conditions).

Demographics and clinical characteristics of the resulting cohort, including age at enrollment, prepregnancy BMI, parity, gestational ages at each visit, and the cross-tabulation of enrollment cohort against pregnancy outcome, are summarized in Table 1. The 23 patients were drawn from three enrollment conditions: Healthy nulliparous women (n=14), women with a personal history of preeclampsia from a previous pregnancy (Prior PE, n=6), and women with polycystic ovarian syndrome (PCOS, n=3; 2 individuals were nulliparous). For conditional generation, we defined three overlapping subgroups based on pregnancy outcome and comorbidity rather than enrollment condition: Uncomplicated (n=18, all patients whose study pregnancy did not result in preeclampsia), PCOS (n=3, a cross-cutting comorbidity; 1 patient also developed PE), and Developed PE (n=5, patients who developed preeclampsia during the study pregnancy, drawn from 2 healthy-enrolled, 2 prior-PE-enrolled, and 1 PCOS-enrolled participants).

5.2 Multiplicity-Weighted Stochastic Attention

We generated synthetic patients using a multiplicity-weighted variant of Stochastic Attention (SA) rooted in modern Hopfield network theory[26]. We stored K memory patterns as columns of a matrix $\hat{\mathbf{X}} \in \mathbb{R}^{d\times K}$ and defined a weighted Hopfield energy landscape:

$$E_r(\boldsymbol{\xi}) = \frac{1}{2}\|\boldsymbol{\xi}\|^2 - \frac{1}{\beta}\log\sum_{k=1}^{K} r_k \exp\left(\beta\,\mathbf{m}_k^\top\boldsymbol{\xi}\right) \tag{1}$$

where $\{\mathbf{m}_k\}_{k=1}^K$ are stored memory patterns, β is an inverse temperature parameter controlling the sharpness of retrieval, and $\{r_k\}_{k=1}^K$ are per-pattern multiplicity weights. When all $r_k = 1$, this reduces to the standard modern Hopfield energy; when $r_k = \rho > 1$ for a designated subset of patterns, the energy landscape is continuously deformed to favor that subset.

This energy determines its induced distribution in closed form. Writing $\pi_r(\boldsymbol{\xi}) \propto \exp[-\beta E_r(\boldsymbol{\xi})]$ and completing the square in each exponent,

$$\exp\left[-\frac{\beta}{2}\|\boldsymbol{\xi}\|^2 + \beta\,\mathbf{m}_k^\top\boldsymbol{\xi}\right] = \exp\left(\frac{\beta}{2}\|\mathbf{m}_k\|^2\right)\exp\left[-\frac{\beta}{2}\|\boldsymbol{\xi}-\mathbf{m}_k\|^2\right], \tag{2}$$

so the target is exactly a finite isotropic Gaussian mixture,

$$\pi_r(\boldsymbol{\xi}) = \sum_{k=1}^{K} q_k\,\mathcal{N}\left(\mathbf{m}_k,\ \beta^{-1}\mathbf{I}\right), \qquad q_k \propto r_k \exp\left(\frac{\beta}{2}\|\mathbf{m}_k\|^2\right). \tag{3}$$

Each stored patient is the center of one component. The common component variance is β^{-1}, and the multiplicity weights enter the mixture probabilities directly. The memories used here are normalized to unit length, so the exponential factor is shared by every component and $q_k = r_k/\sum_j r_j$ exactly. The target can therefore be drawn from ancestrally, by sampling k from q and then $\boldsymbol{\xi} \sim \mathcal{N}(\mathbf{m}_k, \beta^{-1}\mathbf{I})$, with no Markov chain. The cohort analyzed in this work was generated with the Langevin implementation given next, and that is what the reported results describe. We added the analytic sampler afterward as a reference, using the identical normalization, magnitude and decoding pipeline (Table S16).

We sampled from this landscape using an Unadjusted Langevin Algorithm (ULA):

$$\boldsymbol{\xi}_{t+1} = (1-\alpha)\,\boldsymbol{\xi}_t + \alpha\,\hat{\mathbf{X}}\,\mathrm{softmax}\left(\beta\,\hat{\mathbf{X}}^\top\boldsymbol{\xi}_t + \log\mathbf{r}\right) + \sqrt{\frac{2\alpha}{\beta}}\,\boldsymbol{\varepsilon}_t \tag{4}$$

where α is a step size, $\boldsymbol{\varepsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, and $\log\mathbf{r}$ is the element-wise log of the multiplicity vector. The deterministic component drove $\boldsymbol{\xi}_t$ toward a multiplicity-weighted combination of stored patterns, while the stochastic component enabled exploration and novelty. The $\log r_k$ bias in the softmax logits provided a minimal, theoretically grounded mechanism for continuously interpolating between unconditioned generation ($\rho = 1$, all patterns equally weighted) and hard subset curation ($\rho \to \infty$, only designated patterns contribute). This is an inference-time operation that requires no retraining of the energy landscape.

The synthetic cohort analyzed in this work was produced by initializing each Langevin chain near a stored memory pattern and retaining the endpoint of a single trajectory of T=2000 steps. That choice could shape the generator independently of the target distribution, so we characterized it directly. We ran a mixing diagnostic and a four-rung sampler ladder. The ladder compares the published scheme against sphere-initialized, burn-in-controlled, thinned and pooled Metropolis-adjusted sampling, and against exact ancestral draws from Eq. (3). All schemes agree in fidelity and in the memorization behavior relevant to privacy, and the analytic reference agrees with them as well. The published scheme therefore represents the target law faithfully at β^* (Supplementary Fig. S9).

The inverse temperature β controlled the trade-off between retrieval (large β, converging to nearest memory) and generation (small β, uniform mixing). Put plainly, β sets how sharply the sampler commits to one stored patient. When β is small the attention weights are nearly uniform, every stored patient contributes about equally, and samples collapse toward the cohort average. When β is large a single patient dominates and samples reproduce that individual. The useful regime is the transition between the two, where stored patients stop blurring into one average and begin to act as distinct attractors, and β^* is our estimate of where that transition occurs. We identified the critical β^* at the phase transition by locating the inflection of the normalized attention entropy across a logarithmic sweep of β (the entropy definition, sweep parameters and finite-difference procedure are given in the Supplementary Methods). This procedure is domain-agnostic. The phase transition is a geometric property of K patterns in d dimensions, not a property of the features they represent. For our dataset (K=23, d_{PCA}=18), the empirical inflection yielded $\beta^* \approx 2.94$, close to the theoretical prediction $\beta^* = \sqrt{d} \approx 4.24$. A sensitivity analysis across $\beta/\beta^* \in [0.1, 3.0]$ confirmed that the operating point balanced generation novelty against fidelity (Table S1). For weighted sampling, the entropy calculation incorporated the multiplicity bias, yielding a ρ-dependent $\beta^*(\rho)$.

The effective number of patterns contributing to the dynamics was quantified by the participation ratio $K_{\text{eff}} = (\sum_k r_k)^2 / \sum_k r_k^2$, which interpolated smoothly between the total pattern count (unconditioned) and the designated subset size (strongly conditioned). This answers a concrete question. If attention is spread unevenly across the 23 stored patients, how many patients is the generator drawing on? Equal weights give 23, and weights concentrated on three patients give a value near three. Reporting K_{eff} alongside a conditioned cohort states how few patients that cohort rests on. That number bounds what can be concluded from it. For a target effective fraction f_{target} of probability mass on the designated subset, the required multiplicity was $\rho = f_{\text{target}} \cdot K_{\text{bg}}/(K_{\text{des}} \cdot (1 - f_{\text{target}}))$, where K_{des} and K_{bg} are the designated and background pattern counts, respectively.

5.3 Pipeline for Continuous Longitudinal Data

Applying SA to continuous longitudinal clinical data required several adaptations beyond the standard Hopfield sampling framework. We first concatenated each patient's n=72 assay measurements across all three visits into a single vector of dimension $d_{\text{concat}} = 3n = 216$, so that each stored memory pattern encoded a complete longitudinal profile and generated synthetic patients would have internally consistent multi-visit trajectories. We then standardized each feature to zero mean and unit variance and applied PCA retaining 95% of the variance, reducing dimensionality from 216 to d_{PCA}=18. This compression served two purposes: it removed collinearity among the 216 features and yielded a memory-to-dimension ratio of $K/d_{\text{PCA}} \approx 1.28$, placing SA in a favorable operating regime where the number of stored patterns exceeded the dimensionality of the representation.

Standard SA operates on unit-norm patterns, which is appropriate for discrete data (e.g., one-hot protein sequences) but destroys the anisotropic variance structure of continuous clinical measurements, collapsing dispersion across all principal components to a unit sphere. We addressed this with a direction-magnitude decomposition. Before normalization, we recorded each pattern's PCA-space norm $r_k = \|\mathbf{m}_k\|$; we then ran SA on unit-norm patterns to obtain a directional sample $\hat{\boldsymbol{\xi}}$, drew a magnitude r from the empirical distribution of $\{r_k\}$, and reconstructed the rescaled sample as $\boldsymbol{\xi}_{\text{rescaled}} = r \cdot \hat{\boldsymbol{\xi}}$. This preserved the directional structure learned by the Hopfield energy while restoring the natural scale of variation. The rescaled PCA vector was mapped back to the original feature space via inverse PCA and destandardization, then split into three 72-dimensional visit records.

To generate condition-specific cohorts (e.g., PCOS-only), we set the multiplicity $r_k = \rho$ for patterns belonging to the target subgroup and $r_k = 1$ for all others, and sampled using the weighted Langevin update (Eq. 4). The multiplicity ρ was computed to achieve a target effective fraction $f_{\text{target}} = 0.80$ of softmax attention on the designated subset, chosen to strongly bias generation toward the target subgroup while retaining 20% background influence for interpolation diversity, via $\rho = f_{\text{target}} \cdot K_{\text{bg}}/(K_{\text{des}} \cdot (1 - f_{\text{target}}))$. Multiplicity parameters for all three subgroups are listed in Table S2. For the PCOS subgroup (K_{des}=3), this required $\rho \approx 26.7$, reducing the effective pattern count to $K_{\text{eff}} \approx 4.6$. Magnitudes for the direction-magnitude decomposition were drawn from the condition-specific subset of norms, ensuring that the scale of variation matched the target subpopulation rather than the full cohort.

The complete generation pipeline is summarized in Algorithm 1. The full set of SA hyper-parameters used for all experiments is listed in Table S3, with key operating values $\alpha = 0.01$, $T = 2{,}000$ Langevin iterations per sample, and random seed 42 for reproducibility. A control experiment comparing ULA with the Metropolis-Adjusted Langevin Algorithm (MALA) confirmed that the discretization bias is negligible at this step size. Across 10 chains of 5,000 iterations the MALA acceptance rate was $99.9 \pm 0.03\%$, and mean post-burn-in energies and effective sample sizes were indistinguishable between the two samplers (Supplementary Table S8). The pipeline was implemented in Julia (v1.12) using the packages listed in the code repository. Generating 100 synthetic patients required approximately 0.1 seconds on a standard laptop (Apple M-series), comparable to MVN sampling ($\approx$0.15 s), because SA operates in the 18-dimensional PCA space rather than the full 216-dimensional feature space. Each synthetic patient was generated by an independent Langevin chain (no shared state between patients); generation quality was stable across $T \in [500, 4000]$ iterations (median MRE 1.0–1.6%), confirming that T=2,000 was sufficient for convergence.

5.4 Baseline and Validation Framework

As a baseline, we generated synthetic patients by fitting a single multivariate normal (MVN) distribution to the same K=23 concatenated 216-dimensional patient profiles. Critically, both SA and MVN operated on the same concatenated representation. Each patient's three visits were joined into one vector before generation, so that both methods had the opportunity to capture cross-visit dependencies. For SA, this structure was preserved through the Hopfield energy landscape operating on 18-dimensional PCA projections of the concatenated vectors. For MVN, the 216×216 sample covariance matrix had rank at most 22 and was regularized using Ledoit–Wolf shrinkage[21], which inflated eigenvalues in the 194 null dimensions. We then evaluated both SA and MVN synthetic patients through four progressively stringent validation levels. **Level 1 (Marginal Plausibility)** tested whether individual feature distributions matched, using per-feature means, standard deviations, and known physiological relationships. **Level 2 (Joint Structure)** tested whether the cross-visit covariance was preserved, by comparing full 216×216 correlation matrices, eigenvalue spectra, and PCA projections between real, SA, and MVN populations. **Level 3 (Conditional Structure)** tested whether rare subgroups could be amplified while preserving condition-specific signatures, by generating cohorts of 100 patients each from the Uncomplicated (n=18), PCOS (n=3), and Developed PE (n=5) subgroups using multiplicity-weighted sampling. **Level 4 (Mechanistic Consistency)** tested whether synthetic patients produced biologically plausible outputs under an independent mechanistic model. We used the Hockin–Mann BZ2012

Algorithm 1 Multiplicity-Weighted SA for Longitudinal Patient Generation

Require: Patient data $\{\mathbf{x}_k^{(v)}\}$ for $k = 1, \ldots, K$ patients, $v = 1, 2, 3$ visits
Require: Multiplicity vector $\mathbf{r}$ (all ones for unconditioned; $r_k = \rho$ for designated subset)
Ensure: N synthetic longitudinal patient profiles

Concatenate: $\mathbf{x}_k \leftarrow [\mathbf{x}_k^{(1)}; \mathbf{x}_k^{(2)}; \mathbf{x}_k^{(3)}] \in \mathbb{R}^{216}$
Standardize: $\mathbf{z}_k \leftarrow (\mathbf{x}_k - \boldsymbol{\mu}) \oslash \boldsymbol{\sigma}$
PCA: $\mathbf{m}_k \leftarrow \mathbf{W}^\top \mathbf{z}_k \in \mathbb{R}^{18}$ ▷ 95% variance retained
Record norms: $r_k^{\text{mag}} \leftarrow \|\mathbf{m}_k\|$; normalize $\hat{\mathbf{m}}_k \leftarrow \mathbf{m}_k / r_k^{\text{mag}}$
Find β^* via entropy inflection on $\hat{\mathbf{M}} = [\hat{\mathbf{m}}_1, \ldots, \hat{\mathbf{m}}_K]$
for $i = 1, \ldots, N$ **do**
  Initialize $\hat{\boldsymbol{\xi}}_0 \sim \text{Uniform}(\mathcal{S}^{17})$ ▷ random point on unit sphere
  for $t = 0, \ldots, T-1$ **do** ▷ Langevin dynamics
    $\hat{\boldsymbol{\xi}}_{t+1} \leftarrow (1-\alpha)\hat{\boldsymbol{\xi}}_t + \alpha \hat{\mathbf{M}}\,\text{softmax}(\beta^* \hat{\mathbf{M}}^\top \hat{\boldsymbol{\xi}}_t + \log \mathbf{r}) + \sqrt{2\alpha/\beta^*}\,\boldsymbol{\varepsilon}_t$
  end for
  Draw magnitude: $r \sim \{r_k^{\text{mag}}\}$ ▷ condition-specific if weighted
  Rescale: $\mathbf{m}_{\text{synth}} \leftarrow r \cdot \hat{\boldsymbol{\xi}}_T / \|\hat{\boldsymbol{\xi}}_T\|$
  Inverse PCA + destandardize: $\mathbf{x}_{\text{synth}} \leftarrow \boldsymbol{\sigma} \odot (\mathbf{W}\,\mathbf{m}_{\text{synth}} + \bar{\mathbf{z}}) + \boldsymbol{\mu}$
  Split into visit records: $\mathbf{x}_{\text{synth}}^{(1)}, \mathbf{x}_{\text{synth}}^{(2)}, \mathbf{x}_{\text{synth}}^{(3)}$
end for

coagulation model[6,17], a system of 58 ODEs with 64 rate constants, in which 5 rate constants were calibrated on Visit 1 real patients at the population level and the remaining 59 were held at literature values. We ran the model on all real and synthetic patients under two conditions (TF-only and TF+TM) and compared predicted-versus-measured thrombin generation features. The primary metric was cloud overlap: the fraction of synthetic predicted/measured ratios falling within the 5th–95th percentile range of real ratios.

Ethics Statement

This study used de-identified secondary analysis measurements from samples collected longitudinally from women desiring a pregnancy. The primary prospective research study was approved by the Institutional Review Board at the University of Vermont. Written informed consent was obtained from all participants prior to enrollment.

Data Availability

The generated synthetic datasets and all intermediate validation outputs are available in the project repository at `https://github.com/varnerlab/SA-generation-legacy-dataset-paper`. The real patient data were collected under IRB approval at the University of Vermont; de-identified data are available upon reasonable request to the corresponding author.

Code Availability

All code for synthetic patient generation, validation, figure reproduction, and mechanistic model calibration is available at `https://github.com/varnerlab/SA-generation-legacy-dataset-paper`.

The repository includes Julia source code, experiment scripts, and a Python baseline script for CTGAN/TVAE comparison.

Acknowledgments

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Author Contributions

J.D.V. conceived the study, developed the SA pipeline and validation framework, performed computational analyses, and wrote the manuscript. M.C.B. and T.O. oversaw the secondary analysis measurements of coagulation and fibrinolysis markers, dynamic assays, and hormone measurements; provided domain expertise on coagulation biology and thrombin generation assays; and reviewed the manuscript. C.M. was involved in the enrollment of participants in the prospective research study and collection of clinical information of the participants. I.B. designed and oversaw the primary prospective research study of enrolled participants, provided clinical interpretation, and reviewed the manuscript. All authors approved the final version.

Competing Interests

The authors declare no competing interests.

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Table 1: **Demographics and clinical characteristics of the 23 patients with complete longitudinal data across all three visits.** Values are reported as mean ± SD, median (range), or n (%) as appropriate.

Characteristic	Mean ± SD	Median (Range)	n (%)
Age at enrollment (yrs)	30.2 ± 5.1	29 (20–41)	
Race			
White			22 (96)
Not disclosed			1 (4)
Prepregnancy BMI	26.6 ± 5.3	25 (19.0–37.8)	
Parity		0 (0–2)	
Nulliparous			16 (70)
Parous			7 (30)
Enrollment condition			
Healthy nulliparous			14 (61)
PCOS			3 (13)
Prior PE			6 (26)
Cycle day at V1	9.2 ± 3.6	10 (3–14)	
Gestational age at V2 (days)	89.0 ± 5.5		
Gestational age at V3 (days)	217.0 ± 4.0		
Enrollment cohort × pregnancy outcome			
	Uncomplicated	Developed PE	Total
Healthy nulliparous	12	2	14
Prior PE	4	2	6
PCOS	2	1	3
Total	18	5	23

Table 2: Per-visit mean relative error (MRE) between real and SA-generated synthetic patients for representative coagulation features, computed as $|\bar{x}_{\text{synth}} - \bar{x}_{\text{real}}|/|\bar{x}_{\text{real}}|$.

Feature	Visit 1	Visit 2	Visit 3
Factor II (%)	0.018	0.017	0.007
Factor VIII (%)	0.018	0.008	0.005
AT (%)	0.025	0.001	0.008
Fibrinogen (mg/dL)	0.005	0.009	$<$0.001
TF Peak (nM)	0.025	0.002	0.017
TF ETP (nM·min)	0.013	0.008	0.023

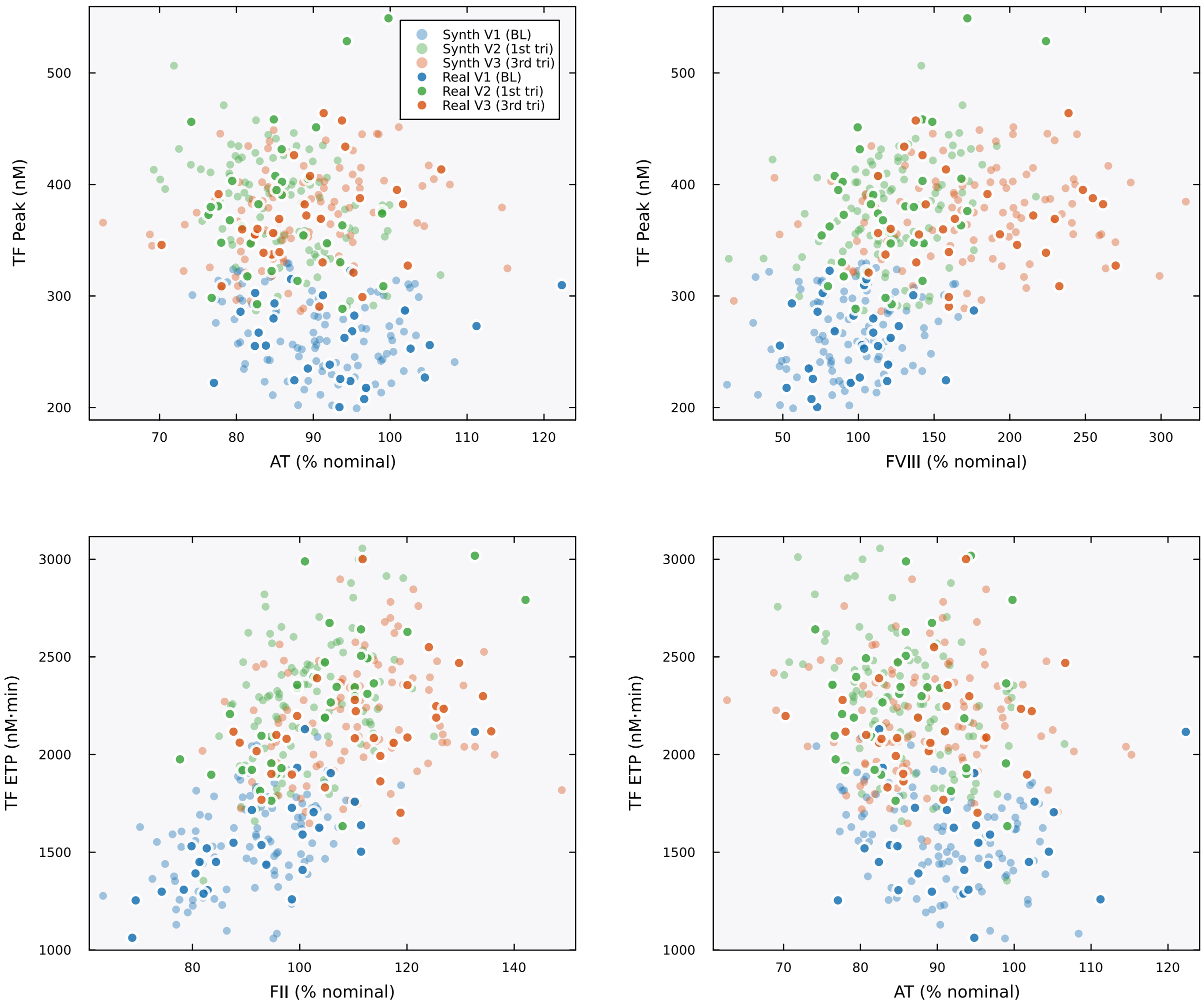

Figure 1: **Physiological correlations are preserved in synthetic patients.** Scatter plots of coagulation factor levels versus thrombin generation parameters across all three visits. Filled markers are real patients; open markers are SA-generated synthetic patients. Colors encode visit: blue (V1/baseline), green (V2/first trimester), orange (V3/third trimester). Real and synthetic patients occupy the same joint regions across all four pairwise relationships, confirming that biologically meaningful dependencies are preserved: AT inversely correlates with TF-initiated peak thrombin (top left), FVIII positively correlates with peak (top right), and both FII (bottom left) and AT (bottom right) show the expected relationships with ETP. The visit-dependent shift in these relationships, reflecting the pregnancy-driven increase in procoagulant factors, is also preserved in the synthetic cohort.

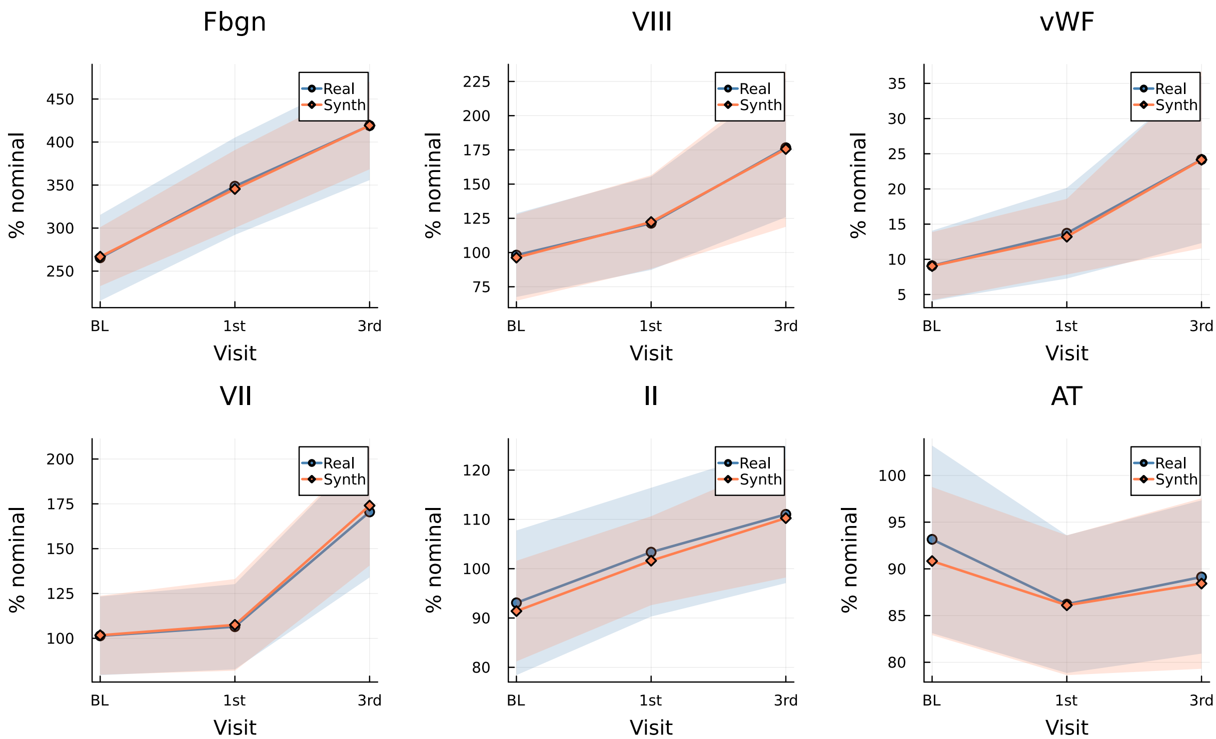

Figure 2: **Pregnancy-driven longitudinal trajectories are reproduced.** Mean ± standard deviation bands for six key coagulation features across visits (BL = baseline, 1st = first trimester, 3rd = third trimester) for real (blue) and SA-generated synthetic (orange) patients. The characteristic pregnancy-driven increases in fibrinogen, Factor VIII, and vWF are captured, as are the stable-to-declining patterns in Factor II and AT.

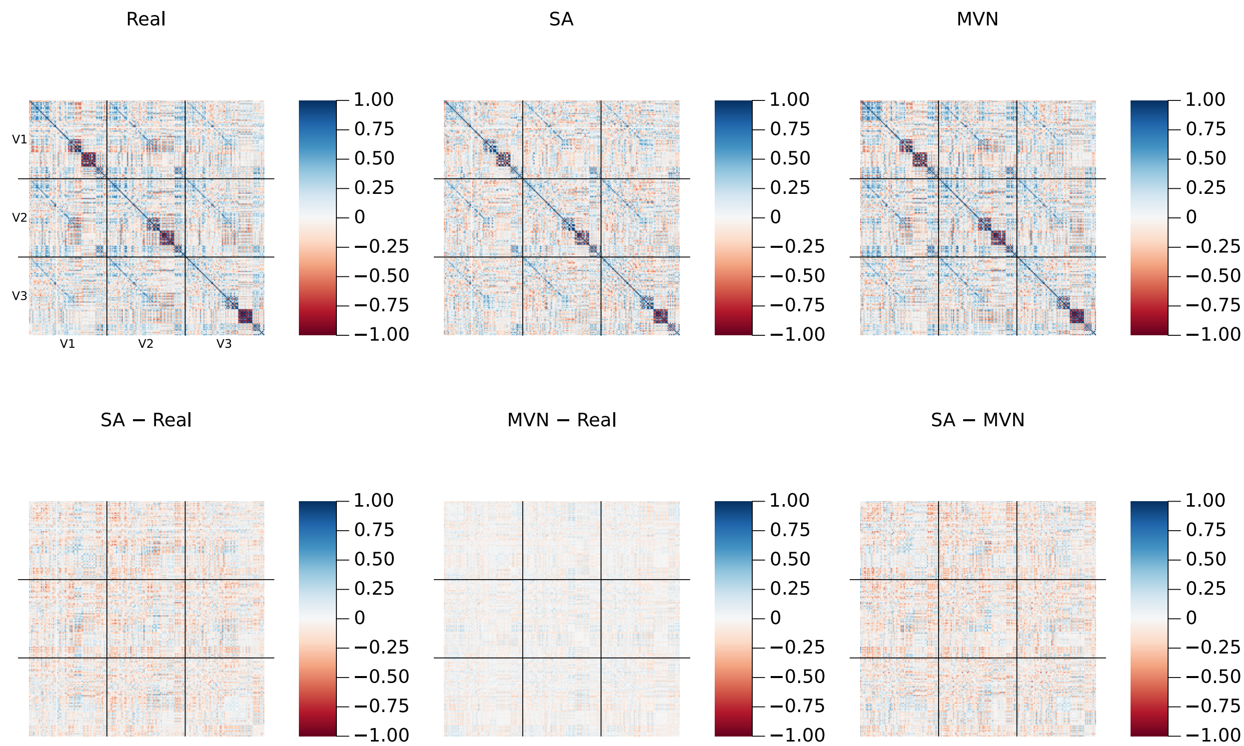

Figure 3: **Cross-visit correlation structure.** Top row: full 216×216 Pearson correlation matrices for Real (K=23, left), SA (N=100, center), and MVN (N=100, right) populations. Each matrix is organized as a 3×3 grid of 72×72 blocks (delineated by black lines), where the diagonal blocks capture within-visit feature correlations (V1–V1, V2–V2, V3–V3) and the off-diagonal blocks capture cross-visit dependencies (e.g., V1–V3: does a patient's baseline Factor VIII predict their third-trimester Factor VIII?). SA preserves both the within-visit and cross-visit block structure. Bottom row: pairwise residual matrices (element-wise subtraction), all plotted on the same ± 1 color scale as the top row. The SA−Real residual shows small, unstructured deviations. The MVN−Real residual is smoother, particularly in the off-diagonal blocks, reflecting the Ledoit–Wolf regularization that systematically shrinks cross-visit correlations toward zero. The SA−MVN residual confirms that the two methods differ most in these off-diagonal blocks, where SA retains longitudinal dependencies that MVN's rank-deficient covariance estimate cannot represent. A version with amplified residual color scale (± 0.5) and Frobenius norms is provided in Fig. S1.

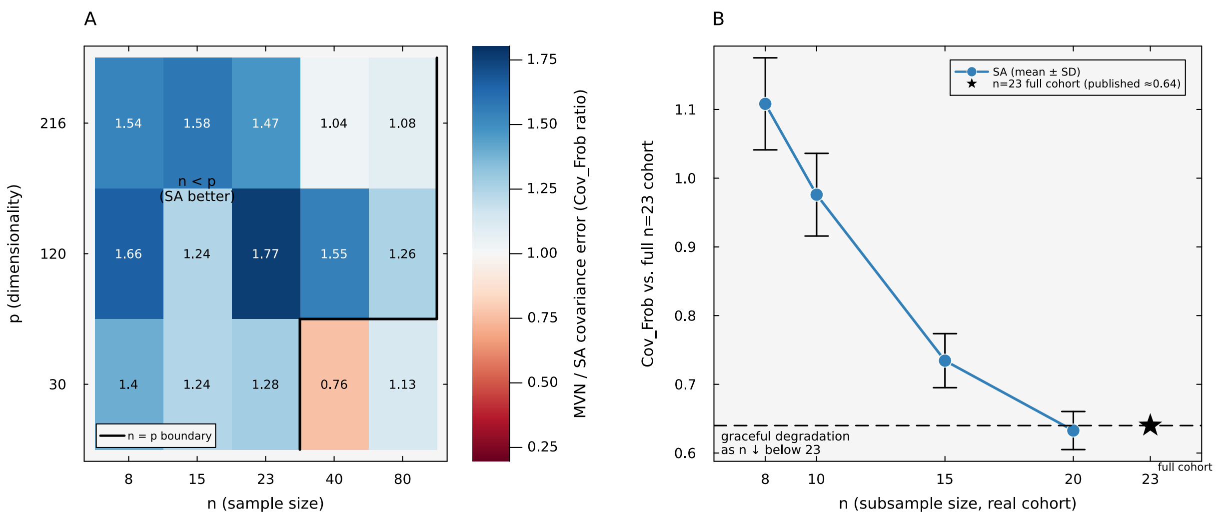

Figure 4: **Recovery across data regimes.** (A) On a low-rank Gaussian ground truth with known population covariance, the ratio of multivariate-Gaussian to SA covariance-recovery error (relative Frobenius distance to the population covariance) across sample size n and dimensionality p, averaged over latent ranks. Values above one (blue) mark cells where SA recovers the population covariance more accurately than a multivariate Gaussian. The advantage is largest in the rank-deficient $n < p$ regime that contains the real cohort's shape (n=23, p=216), and narrows as n approaches p, where the Gaussian is well specified. The target covariance is fixed in advance, so this measures generalization and not reproduction of the training draw. (B) On the real cohort, SA's recovery of the full 23-patient cross-visit correlation structure as the cohort is subsampled to n=20, 15, 10, 8 (mean and standard deviation over ten random subsets, black bars). Recovery degrades gracefully toward the full-cohort value (star) instead of collapsing.

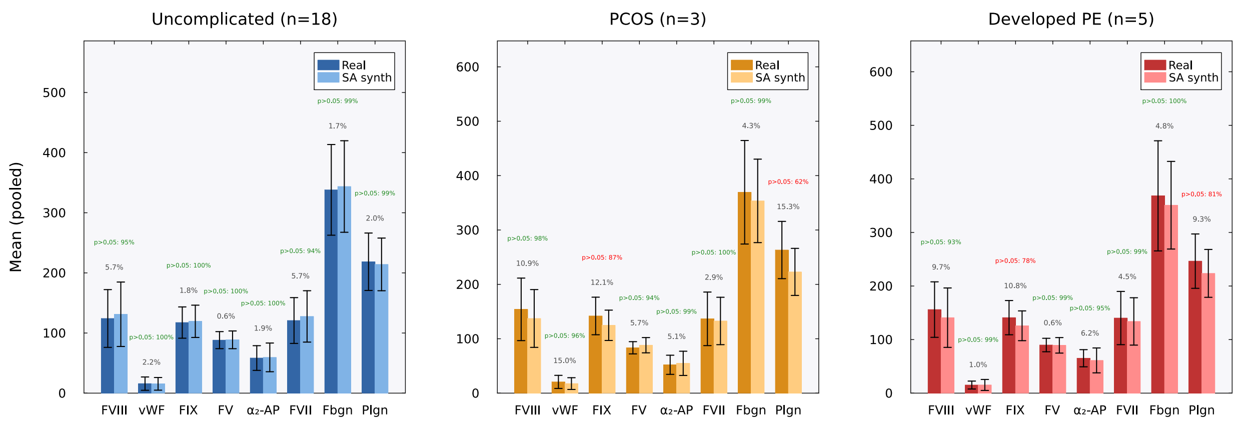

Figure 5: **Condition-specific feature preservation.** Grouped bar charts comparing real (dark bars) and SA-generated synthetic (light bars) patient means ($\pm$ SD error bars) for eight coagulation features, shown separately for each clinical subgroup: Uncomplicated (n=18, left, blue), PCOS (n=3, center, orange), and Developed PE (n=5, right, red). Each bar is the mean of that assay pooled across all three visits (V1–V3); factor activities (FVIII, vWF, FIX, FV, α_2-AP, FVII, plasminogen) are expressed as % of normal pooled reference plasma, and fibrinogen (Fbgn) in mg/dL. Gray percentages indicate the mean relative error (MRE). Green annotations show the bootstrap Mann–Whitney equivalence result: the fraction of 1,000 bootstrap replicates (synthetic subsampled to n_{real}) in which $p > 0.05$, i.e., real and synthetic could not be distinguished. Features achieving $\geq$90% are shown in green; <90% in red. Across all 24 feature–condition pairs, 20 (83%) were statistically indistinguishable in $\geq$90% of replicates (median: 98.6%). The four pairs below this threshold were FIX and plasminogen in the PCOS and Developed PE subgroups, where small subgroup sizes and large inter-patient variability reduced test power.

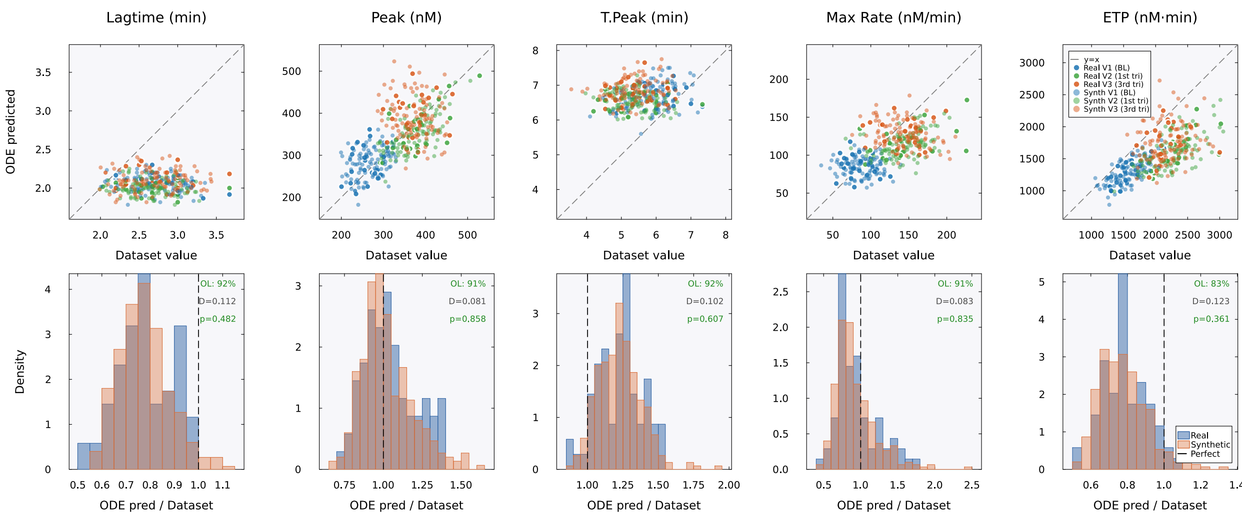

Figure 6: **Mechanistic validation (TF-only).** Top row: BZ2012 ODE-predicted TGA values (vertical axis) versus dataset TGA values (horizontal axis) for five thrombin generation parameters. The dataset values are the TGA measurements from each patient's record, present for both real and synthetic patients. The ODE-predicted values are computed by running each patient's coagulation factor levels through the 58-species BZ2012 model. The dashed line indicates where the ODE model would perfectly reproduce the dataset value (y=x); systematic deviations from this line (e.g., lagtime underprediction) reflect ODE model bias, not synthetic data failure. Filled markers are real patients, open markers are SA-generated synthetic patients, colored by visit: blue (V1/baseline, training set), green (V2/first trimester), orange (V3/third trimester). Real and synthetic patients occupy the same cloud with the same pattern of ODE model bias. Bottom row: ODE-predicted/dataset ratio distributions for real (blue) and synthetic (orange) patients. If SA had generated biologically implausible factor combinations, the ODE model would process them differently, producing divergent ratio distributions. Instead, the distributions overlap substantially. Each panel is annotated with cloud overlap (OL), the two-sample Kolmogorov–Smirnov statistic (D), and p-value. All five features show $p > 0.35$, confirming that the ODE model cannot distinguish real from synthetic patients.

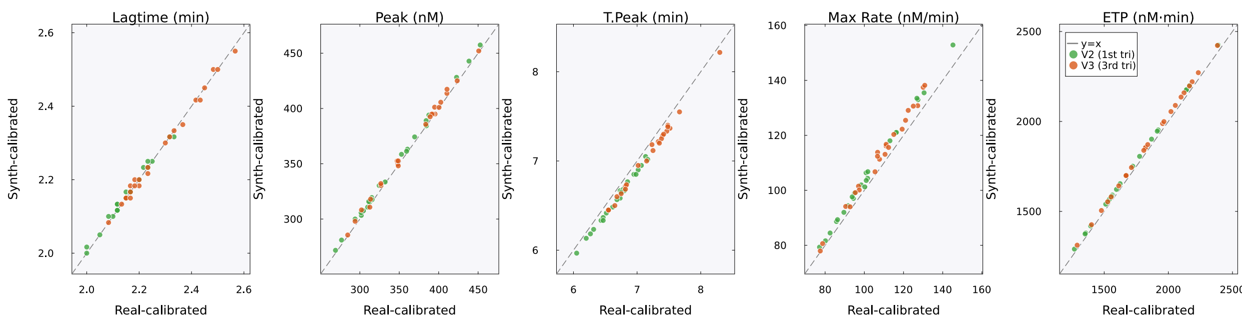

Figure 7: **Downstream utility: synth-calibrated vs. real-calibrated mechanistic model predictions on held-out real patients.** Each panel compares the BZ2012 ODE predictions for one TGA feature when the model is calibrated on real V1 patients (horizontal axis) versus synthetic V1 patients (vertical axis), evaluated on the same held-out real V2 and V3 patients. Points near the $y{=}x$ line indicate that the two calibrations produce equivalent predictions. Green: V2 (first trimester); orange: V3 (third trimester). The synth-calibrated model matched or outperformed the real-calibrated model on all five features (overall ratio 0.94×), demonstrating that a mechanistic model trained entirely on synthetic data generalizes to real patients as well as one trained on real data. Both calibrations used 11 random restarts; all starts converged.

Supplementary Information

Supplementary Tables

Table S1: **Inverse temperature sensitivity analysis.** SA generation quality as a function of β/β^*, where $\beta^* \approx 2.94$ was identified via entropy inflection in the 18-dimensional PCA space (K=23 stored patterns). Operating at $\beta = \beta^*$ (bold row) balances generation novelty against fidelity. Med MRE = median relative error of per-feature means; Novelty = $1 - \max_k \cos(\hat{\xi}, \hat{m}_k)$.

β/β^*	β	Noise scale	PC1 std ratio	Novelty	Med MRE (%)
0.1	0.29	0.261	0.538	0.542	0.7
0.3	0.88	0.151	0.550	0.536	0.7
0.5	1.47	0.117	0.562	0.528	0.7
1.0	**2.94**	**0.082**	**0.589**	**0.502**	**0.6**
1.5	4.41	0.067	0.613	0.474	0.7
2.0	5.88	0.058	0.638	0.446	0.7
3.0	8.82	0.048	0.689	0.403	0.8

Table S2: **Multiplicity weighting parameters for conditional generation.** The multiplicity ratio ρ was computed to achieve a target effective fraction $f_{\text{target}} = 0.80$ of softmax attention on the designated patterns. K_{eff} is the participation ratio quantifying the effective number of patterns contributing to generation.

Condition	K_{sub}	K_{bg}	ρ	f_{eff}	K_{eff}
Uncomplicated	18	5	1.11	0.80	23.0
PCOS	3	20	26.67	0.80	4.6
Developed PE	5	18	14.40	0.80	7.7

Table S3: **SA generation hyperparameters.** The same values were used for unconditioned and conditioned generation except where noted.

Parameter	Value	Description
α	0.01	Langevin step size (see Table S8)
T	2,000	Langevin iterations per sample
β^*	2.94	Inverse temperature (entropy inflection)
PCA threshold	95%	Variance retained
d_{PCA}	18	PCA dimensionality
N	100	Synthetic patients per cohort
Random seed	42	For reproducibility
f_{target}	0.80	Target attention fraction (conditioned only)

Table S4: **BZ2012 ODE model calibration.** Five rate constants were calibrated on Visit 1 real patients (n=23) using bounded Nelder–Mead optimization in log-scale-factor space. The cost function was the mean normalized relative squared error across five TGA features under TF-only conditions. Bounds: $\pm\log(100)$. Convergence: $f_{\text{tol}} = 10^{-6}$, 500 iterations, 11 random restarts.

Rate constant	Role	Default	Calibrated	Scale
prothrombinase k_{cat}	Thrombin burst	63.5 s^{-1}	21.94 s^{-1}	0.35×
intrinsic Xase k_{cat}	Amplification Xa	8.2 s^{-1}	0.169 s^{-1}	0.021×
extrinsic Xase k_{cat}	Initiation Xa	6.0 s^{-1}	93.2 s^{-1}	15.5×
PC activation k_{cat}	Protein C activation	0.41 s^{-1}	0.890 s^{-1}	2.17×
mIIa conversion k	mIIa→IIa	2.3×10^{8}	1.15×10^{9}	5.01×

Units for mIIa conversion: $\text{M}^{-1}\text{s}^{-1}$

Table S5: **Summary of mean relative errors across all 72 features.** Per-visit distribution of MREs between real and SA-generated synthetic patient means. The full feature-by-feature table (72 features × 3 visits = 216 entries) is provided in the code repository as `paper_summary_statistics.csv`.

Visit	Median MRE	Mean MRE	25th pctl	75th pctl	Max MRE	Below 5%
V1 (baseline)	0.022	0.029	0.008	0.039	0.158	59/72 (81.9%)
V2 (1st tri)	0.009	0.021	0.004	0.022	0.224	65/72 (90.3%)
V3 (3rd tri)	0.011	0.016	0.005	0.024	0.062	69/72 (95.8%)
Pooled (all 3 visits)	**0.012**	**0.022**	**0.005**	**0.028**	**0.224**	**193/216 (89.4%)**

Pooled median MRE 95% bootstrap CI: (0.98%, 1.60%) over 10,000 resamples (seed 42).

Table S6: **Comparison with deep generative baselines.** CTGAN and TVAE were trained on the same 90 per-visit records (23 patients $\times$ 3 visits, 72 features) at three epoch counts. SA and MVN results from the main text are shown for reference. CTGAN fails at this sample size (median MRE $\approx$ 19%), consistent with GAN mode collapse on small datasets. TVAE achieves comparable marginal fidelity at high epoch counts but cannot capture cross-visit covariance or perform conditional generation.

Method	Epochs	Med MRE	Med KS	Corr MAE
SA	–	0.019	0.169	0.200
MVN	–	0.024	0.161	0.090
CTGAN	300	0.189	0.364	0.261
CTGAN	1,000	0.195	0.349	0.265
CTGAN	3,000	0.187	0.341	0.180
TVAE	300	0.037	0.174	0.152
TVAE	1,000	0.026	0.248	0.082
TVAE	3,000	0.018	0.224	0.092

Table S7: **PCA variance threshold sensitivity analysis.** SA generation quality is stable across thresholds from 85% to 99%. Median MRE remains between 1.0% and 1.7% across all thresholds. The 95% threshold used in the main text (bold) is not a critical design choice.

Threshold	d_{PCA}	K/d	β^*	Novelty	Med MRE	Corr MAE
85%	13	1.77	2.94	0.420	0.0098	0.124
90%	15	1.53	2.94	0.477	0.0160	0.126
95%	**18**	**1.28**	**2.94**	**0.500**	**0.0124**	**0.143**
97.5%	20	1.15	2.94	0.543	0.0120	0.135
99%	21	1.10	2.94	0.541	0.0172	0.138

Table S8: **ULA vs MALA sampling diagnostics.** Ten chains of 5,000 iterations each were run with both the Unadjusted Langevin Algorithm (ULA) and the Metropolis-Adjusted Langevin Algorithm (MALA) at step size $\alpha = 0.01$ and $\beta^* = 2.94$ on the 18-dimensional PCA memory (K=23 patterns). The MALA acceptance rate of 99.9% confirms that virtually every ULA proposal satisfies detailed balance, and the indistinguishable energy and effective sample size statistics confirm negligible discretization bias. This comparison bounds the discretization error of the reported sampler. The direct reference for the target itself is the analytic sampler of Eq. (3), introduced below.

Metric	ULA	MALA
Acceptance rate (%)	– (no rejection)	99.9 ± 0.03
Mean energy (post-burn-in)	1.932 ± 0.141	1.886 ± 0.231
τ_{int}	109.5 ± 49.5	106.7 ± 30.4
Effective sample size	41.8 ± 14.2	40.0 ± 10.2

Burn-in: 1,000 iterations discarded. τ_{int}: integrated autocorrelation time estimated with a windowed estimator (threshold 0.05). ESS = $(T - T_{\mathrm{burn}})/\tau_{\mathrm{int}}$.

Table S9: **Memorization diagnostics.** Two complementary checks that SA-generated patients are not memorized copies of the K=23 stored profiles. Novelty score is the angular distance between a synthetic sample and its nearest stored pattern at $\beta = \beta^*$. Nearest-neighbor distances are computed in the per-visit standardized 72-dimensional feature space.

Metric	Value
Mean novelty score, $1 - \max_k \cos(\hat{\xi}, \hat{m}_k)$	0.50
Median synth–real nearest-neighbor distance (std. units)	6.14
Median real–real nearest-neighbor distance (std. units)	6.87
Distance ratio (synth–real / real–real)	0.89

Table S10: **Bootstrap Mann–Whitney equivalence per feature and condition.** For each of 24 feature–condition pairs, we subsampled the synthetic cohort to match the real sample size, applied a Mann–Whitney U test, and repeated 1,000 times. Frac. $p > 0.05$ is the fraction of replicates in which the test could not distinguish the synthetic from the real distribution. 20 of 24 pairs achieve $p > 0.05$ in $\geq 90\%$ of replicates; the four failing pairs (italicized) all involve high-variance features in the smallest subgroups.

Condition	Feature	MRE	Frac. $p > 0.05$
Uncomplicated (n=18)	FVIII	0.057	0.952
Uncomplicated (n=18)	vWF	0.022	0.999
Uncomplicated (n=18)	FIX	0.018	0.998
Uncomplicated (n=18)	FV	0.006	0.996
Uncomplicated (n=18)	α_2-AP	0.019	1.000
Uncomplicated (n=18)	FVII	0.057	0.941
Uncomplicated (n=18)	Fbgn	0.017	0.987
Uncomplicated (n=18)	Plgn	0.020	0.988
PCOS (n=3)	FVIII	0.109	0.985
PCOS (n=3)	vWF	0.150	0.959
PCOS (n=3)	*FIX*	*0.121*	*0.873*
PCOS (n=3)	FV	0.057	0.944
PCOS (n=3)	α_2-AP	0.051	0.987
PCOS (n=3)	FVII	0.029	0.999
PCOS (n=3)	Fbgn	0.043	0.992
PCOS (n=3)	*Plgn*	*0.153*	*0.615*
Developed PE (n=5)	FVIII	0.097	0.932
Developed PE (n=5)	vWF	0.010	0.986
Developed PE (n=5)	*FIX*	*0.108*	*0.780*
Developed PE (n=5)	FV	0.006	0.987
Developed PE (n=5)	α_2-AP	0.062	0.950
Developed PE (n=5)	FVII	0.045	0.993
Developed PE (n=5)	Fbgn	0.048	0.996
Developed PE (n=5)	*Plgn*	*0.093*	*0.814*

Median Frac. $p > 0.05$ across all 24 pairs: 0.986. Failing pairs (< 0.90, italicized): PCOS–FIX, PCOS–Plgn, DPE–FIX, DPE–Plgn.

Table S11: **Mechanistic validation diagnostics per feature and condition.** Cloud overlap is the fraction of synthetic ODE-predicted/measured ratios falling within the 5th–95th percentile of the corresponding real distribution. KS D and KS p are from a two-sample Kolmogorov–Smirnov test on the same ratio distributions. Real n=69 (23 patients × 3 visits); synthetic n=300 (100 patients × 3 visits).

Condition	TGA Feature	Cloud overlap	KS D	KS p
TF-only	Lagtime	0.92	0.112	0.48
TF-only	Peak	0.91	0.081	0.86
TF-only	T.Peak	0.92	0.102	0.61
TF-only	Max Rate	0.91	0.083	0.84
TF-only	ETP	0.83	0.123	0.36
TF+TM	Lagtime	0.92	0.127	0.32
TF+TM	Peak	0.88	0.079	0.88
TF+TM	T.Peak	0.91	0.110	0.51
TF+TM	Max Rate	0.88	0.096	0.67
TF+TM	ETP	0.89	0.112	0.48

TF-only cloud overlap range: 0.83–0.92. TF+TM cloud overlap range: 0.88–0.92. Pooled across both conditions and all five features, 0.902. KS D range across both conditions: 0.079–0.127, all $p > 0.30$.

Table S12: **Downstream utility — per-feature held-out V2/V3 performance.** Median relative error of the BZ2012 ODE-predicted TGA values against the held-out real V2 and V3 measurements, for the model calibrated on real V1 patients (K=23) and the model calibrated on synthetic V1 patients (N=100). The synth-calibrated model achieves slightly lower error on every feature. Synth/Real ratio < 1 favors the synth-calibrated model.

TGA Feature	Real-cal MRE	Synth-cal MRE	Synth/Real
Lagtime	0.165	0.162	0.98×
Peak	0.097	0.087	0.90×
T.Peak	0.330	0.306	0.93×
Max Rate	0.286	0.259	0.91×
ETP	0.197	0.183	0.93×
Overall median	**0.201**	**0.189**	**0.94×**

Table S13: **PCA-space ablation.** Five generators fit in the identical d=18 PCA subspace and evaluated on the same harness as SA. Median relative error (MRE) and the mechanistic cloud overlap measure fidelity. Median novelty measures distance from the stored patterns, and Frac. novel is the fraction above the 0.2 novelty threshold. The cross-visit Frobenius error is the deviation of the synthetic correlation matrix from the real one. A lower value means staying closer to the 23 real patients, so this metric rewards memorization. The shared subspace is necessary but not sufficient. k-nearest-neighbor interpolation collapses novelty, and the kernel-density estimator degrades marginal and mechanistic fidelity, despite both being fit in it. The mixture and copula baselines match SA on the fidelity axes.

Method	MRE (%)	Cross-visit Frob.	Mech. overlap	Median novelty	Frac. novel
SA	1.24	0.641	0.902	0.514	1.00
PCA+GMM	1.13	0.422	0.918	0.497	0.94
PCA+KDE	1.94	0.337	0.823	0.249	0.65
PCA+kNN	1.63	0.587	0.938	0.051	0.05
PCA+Copula	1.34	0.417	0.934	0.462	1.00

Table S14: **Convex-hull membership does not discriminate at K=23 in d=18.** Hull membership and distance for the 100 synthetic patients against the 23 real patients, and for each real patient against the other 22. The radial factor t^* is where the ray from the hull centroid through a point exits the hull, so $1/t^*$ is the factor by which a point overshoots the boundary and $t^* \geq 1$ would mean the point is inside. Distances and overshoots are severity-matched, with each point's nearest vertex removed on both sides, because a held-out real patient necessarily loses its own vertex. Matched this way, the distance to the hull is not distinguishable between synthetic and real patients (Mann–Whitney $p = 0.07$), while the radial overshoot is smaller for the synthetic ones ($p < 10^{-4}$). Both tests treat the 100 synthetic patients as independent, which they are not, so the reported significance is optimistic. The geometry rows show why no point of either kind falls inside. The hull attains the patients' typical radius only along the 23 vertex directions, and the sampler returns every sample to the unit sphere. A point inside the hull of unit-norm patterns would have to be one of those patterns.

Quantity	Synthetic (n=100)	Real, leave-one-out (n=23)
Fraction inside hull	0.00	0.00
Median distance to hull	11.0	11.9
Median radial overshoot $1/t^*$	10.5	15.2
Hull geometry of the real cohort		
Median patient radius	13.8	
Median hull extent, generic direction	2.0	
Largest t^* attained on the unit sphere	0.231	
Plausibility of the out-of-hull synthetic patients		
Biological constraints satisfied	81/100	
Mechanistic band, per feature	83 to 92%	
Mechanistic band, all features and visits	41/100	

Table S15: **Membership inference recovers memory membership, not unseen patients.** Median nearest-neighbor distances in the standardized 216-dimensional space, and the attack's discrimination. The attack retrains the generator on 15 training patients and scores each real record by its distance to the resulting synthetic set, treating the 15 training patients as members and the 8 held-out patients as non-members. Every quantity comes from the same protocol. We drew 40 partitions at random with the generation seed held fixed, so that only the partition varies, and pooled the distances across all partitions. Non-members sit at the same distance from the synthetic cohort as unrelated real patients sit from one another. The synthetic cohort carries no proximity signal about patients the generator never saw, and the discrimination comes entirely from members sitting closer.

Quantity	Value
Median nearest-neighbor distance	
Training member to nearest synthetic	10.7
Held-out non-member to nearest synthetic	15.4
Real to nearest real, leave-one-out	15.9
Synthetic to nearest real	13.8
Membership-inference AUC over 40 *random splits*	
Mean	0.97
Standard deviation	0.02
Median	0.98
Minimum	0.90
Maximum	1.00
Splits reaching perfect separation	9/40
Splits with AUC above 0.9	39/40

Supplementary Figures

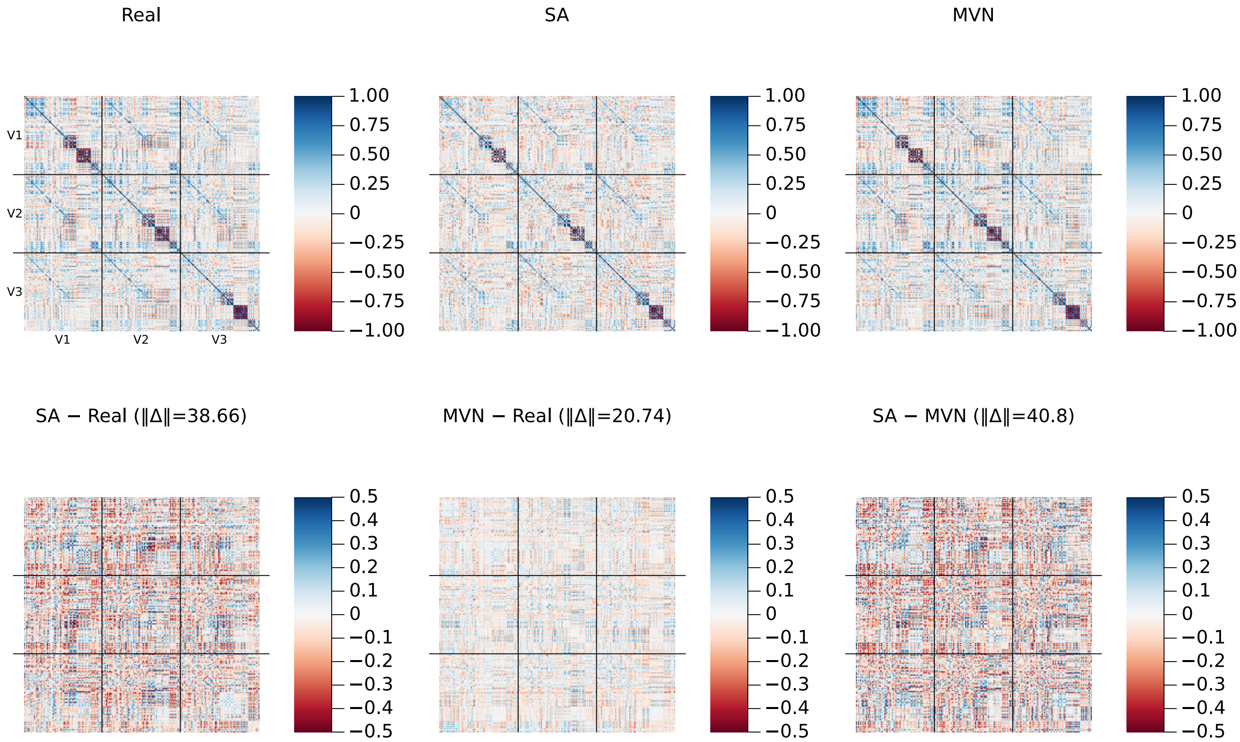

Figure S1: **Cross-visit correlation structure (amplified residual scale).** Same layout as the main-text correlation figure, but with the bottom-row residual panels plotted on a ± 0.5 color scale (vs. ± 1.0 in the main text) to reveal fine structure. Frobenius norms of each residual matrix are shown in the panel titles.

Supplementary Methods: locating the critical inverse temperature

The operating point β^* was located from the attention entropy. For a candidate β we computed the Shannon entropy of the attention weights $\mathrm{softmax}(\beta\, \hat{\mathbf{X}}^\top \boldsymbol{\xi})$ at 20 probe points placed at stored patterns, averaged those entropies, and normalized by $\log K$. The normalized curve runs from 0 to 1. A value of 0 means one pattern carries all the weight, and a value of 1 means every pattern carries equal weight. Sweeping β logarithmically over 80 values from 0.1 to 1000 traces this curve from the diffuse regime into the concentrated one. We took β^* to be the point of most negative second derivative of the normalized entropy with respect to $\log \beta$, computed by finite differences over the sweep. That point is the sharpest bend in the curve, and it marks where attention stops being shared across patterns and begins to localize on one. For multiplicity-weighted sampling we repeated the same computation with the $\log r_k$ bias included in the softmax logits. This gives a ρ-dependent $\beta^*(\rho)$, because the weighting changes how quickly attention concentrates as β grows. For random unit-norm patterns the transition is predicted at $\beta^* = \sqrt{d}$, the theoretical reference quoted in the main text.

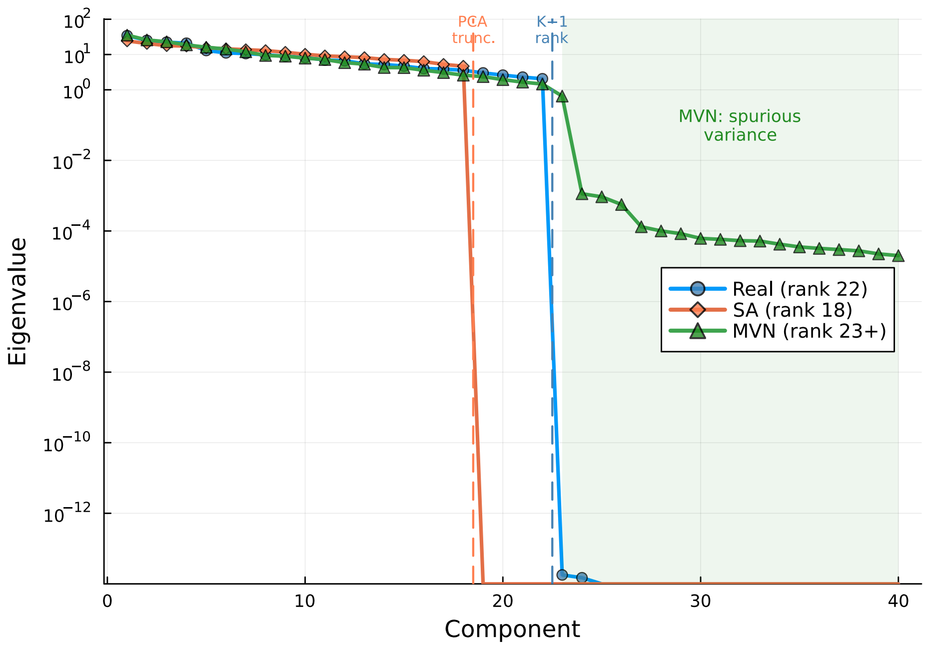

Figure S2: **Covariance eigenvalue spectrum.** Log-scale eigenvalues of the 216 × 216 sample covariance for Real (K=23, rank 22), SA (N=100, rank 18), and MVN (N=100) populations. Dashed vertical lines mark the SA PCA truncation point (component 18) and the data rank limit (component 22). MVN's Ledoit–Wolf regularization inflates eigenvalues beyond component 22 (green shaded region), introducing spurious variance in 194 dimensions where the data contain no signal.

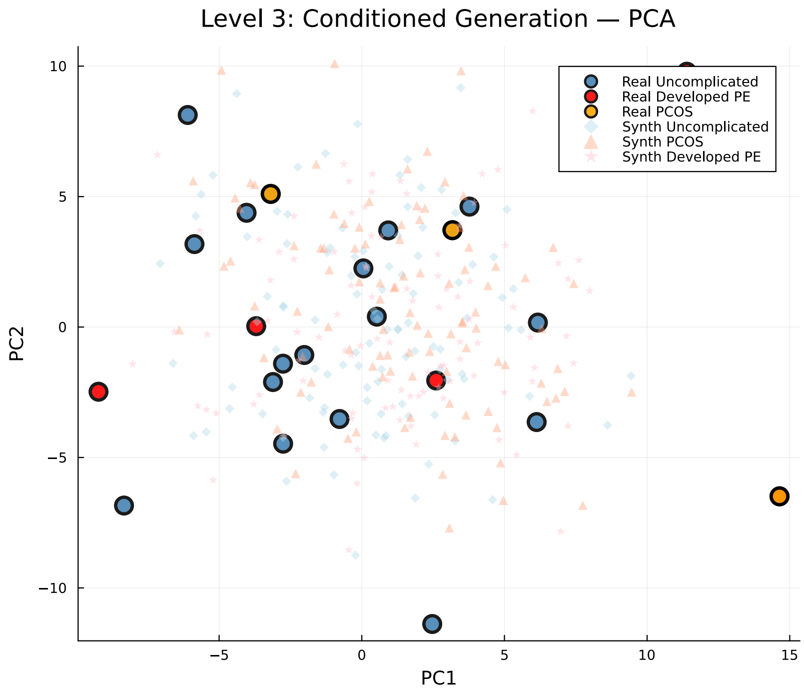

Figure S3: **Conditioned generation: PCA projection (pooled).** Large markers: real patients (blue = Uncomplicated, n=18; orange = PCOS, n=3; red = Developed PE, n=5). Small markers: synthetic patients (N=100 per condition). With only 3 PCOS and 5 Developed PE patients, the subgroups do not form clearly separable clusters in the first two principal components, but the synthetic cohorts concentrate around their respective real counterparts.

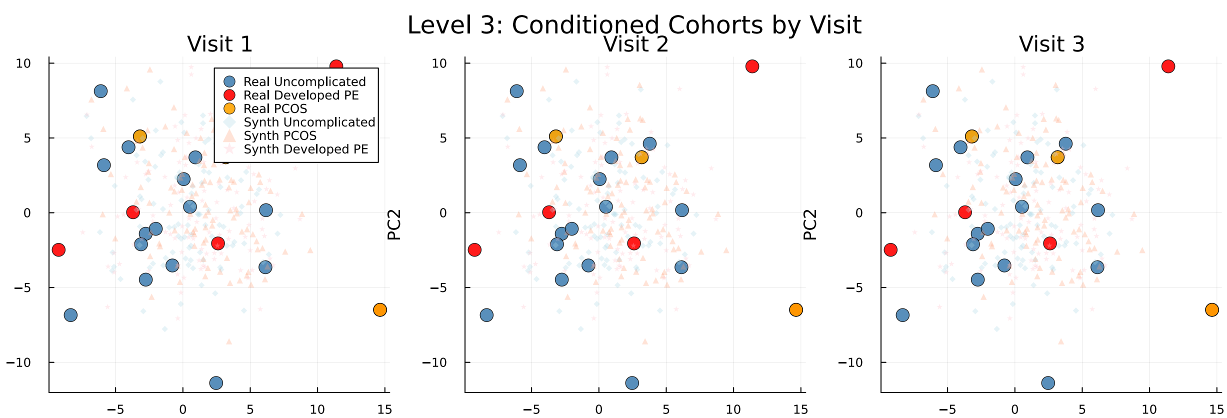

Figure S4: **Conditioned generation: PCA projection by visit.** Same as Fig. S3 but separated by visit (V1, V2, V3). Condition-specific clustering is maintained within each individual visit.

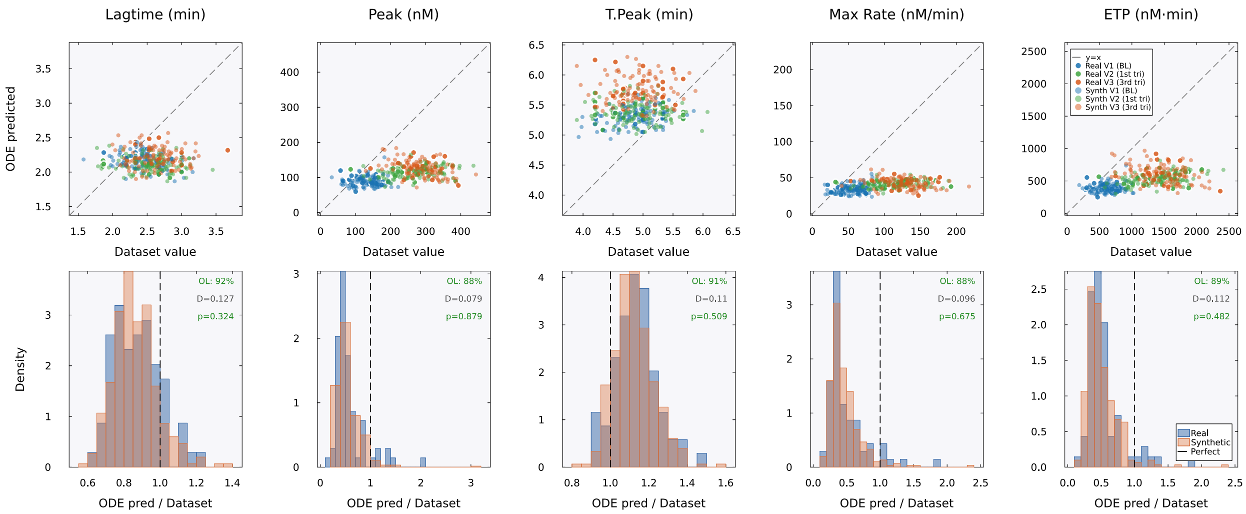

Figure S5: **Mechanistic validation (TF+TM condition).** Same layout as main-text Fig. 6 but under TF+TM conditions. The model systematically underpredicts peak and maximum rate ($\approx 0.5\times$ measured) for both real and synthetic patients, reflecting overcorrection of the protein C anticoagulant pathway. Cloud overlap remains high at 88–92%, confirming that the model processes real and synthetic patients identically even under imperfect calibration conditions.

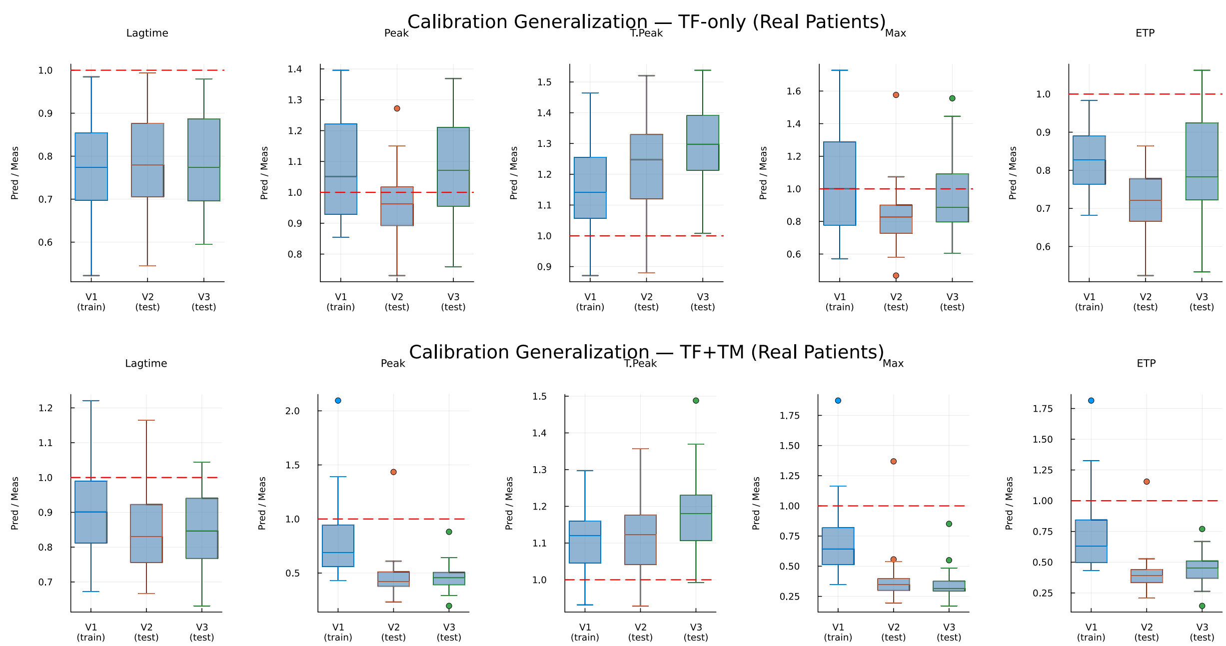

Figure S6: **Cross-visit calibration generalization.** Boxplots of the ODE-predicted/dataset ratio by visit for real patients. The model was calibrated on Visit 1 only. Top: TF-only conditions, where ratios near 1.0 at Visits 2 and 3 indicate that the calibration generalizes across pregnancy timepoints. Bottom: TF+TM conditions, where the calibration does not generalize as well, particularly for peak and maximum rate.

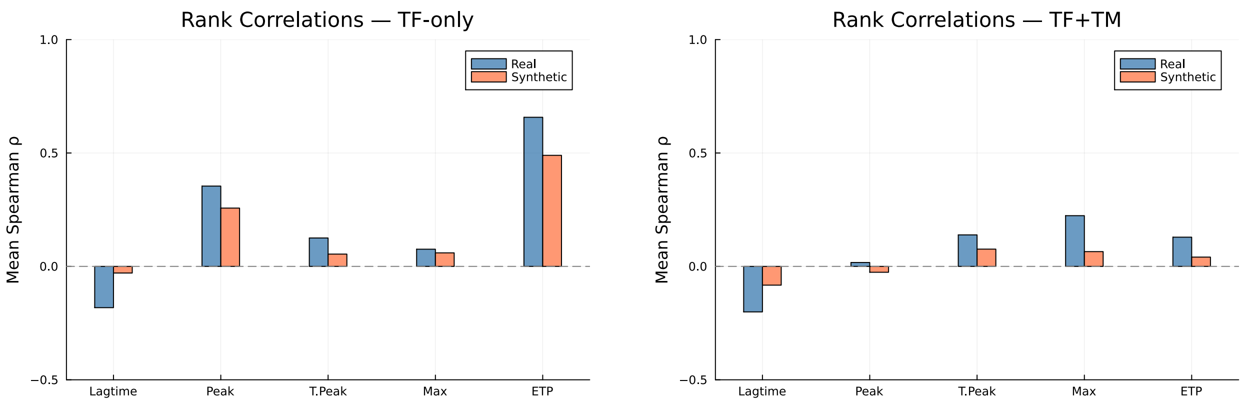

Figure S7: **Spearman rank correlations between ODE-predicted and dataset TGA features.** Left: TF-only. Right: TF+TM. ETP is the best-predicted feature ($\rho \approx 0.6$–0.8 for real patients under TF-only). Weak rank correlations for other features are expected given the 5-parameter population-level calibration.

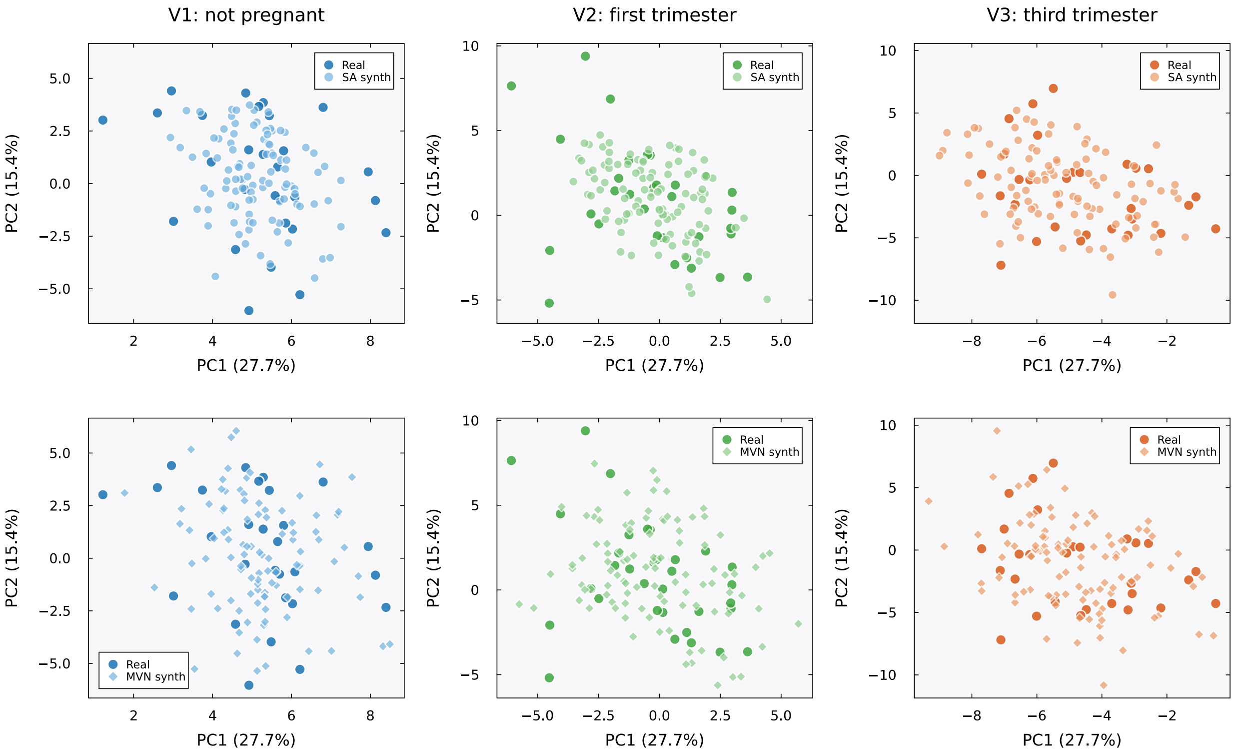

Figure S8: **PCA projections by visit: SA vs. MVN.** Each panel shows the first two principal components (PC1: 27.7% variance, PC2: 15.4%) computed from standardized per-visit real patient data (K=23 patients per visit, 72 features). Dark markers indicate real patients; lighter markers indicate synthetic patients (N=100). Within each visit (column), the SA panel (top) and MVN panel (bottom) share identical axis ranges, so the relative dispersion of the two methods can be read against the same real data cloud. Top row: SA-generated patients (circles) cluster tightly around the real data cloud at all three visits. Bottom row: MVN-generated patients (diamonds) show greater dispersion, particularly at Visits 2 (first trimester) and 3 (third trimester), where MVN patients extend beyond the convex hull of the real data. This excess scatter reflects the spurious variance introduced by Ledoit–Wolf regularization of the rank-deficient covariance (Fig. S2). Colors encode visit: blue (V1, not pregnant), green (V2, first trimester), orange (V3, third trimester).

Sampling correctness and privacy diagnostics

We assessed re-identification risk directly. For every synthetic patient we computed the distance to the closest real record (DCR) in the standardized 216-dimensional space and compared the synthetic-to-real distribution against the leave-one-out real-to-real distribution. Synthetic patients sit slightly closer to the real cohort than the real patients sit to one another, with median DCR 13.8 against 15.9. We also ran a nearest-neighbor membership-inference attack, which scores each real record by its proximity to the synthetic set over repeated random train/holdout splits. It separated the 15 training patients from the 8 held-out patients with a mean AUC of 0.97 (standard deviation 0.02 over 40 splits). At the published operating point the generator therefore reproduces its training cohort closely enough to be identifiable.

To attribute this behavior we asked whether it reflects the target distribution $\pi(\xi) \propto \exp(-\beta E)$ or merely the sampler used to draw from it. A mixing diagnostic showed that at β^* the target is multimodal, with one component centered on each of the K=23 stored patterns. It also showed that discretization bias in the Langevin updates is negligible, with Metropolis acceptance near 0.999 (Supplementary Fig. S10). We then constructed a four-rung ladder (Table S16, Fig. S9) that holds β, the memory, and the decoding fixed and varies only the sampler. The rungs run from the published anchor-initialized single-endpoint scheme, to sphere initialization, to burn-in-controlled thinned pooling, to Metropolis-adjusted sampling. Across the ladder the membership-inference AUC stays near 0.96 to 0.97. The median DCR rises only from 14.0 to 14.3 and never reaches the real-to-real baseline of 15.9, at a modest cost in marginal fidelity. Proper sampling of π does not remove the memorization. At K=23 the memorization is a property of the distribution.

The remaining lever is the inverse temperature. Lowering β spreads probability mass away from the stored patterns and should improve privacy, at the expense of the memorization that also underlies fidelity. We swept β across a 40-fold range around β^* (Fig. S9C). The sweep confirmed the trade-off and bounded it sharply. The median DCR rises monotonically as β falls, but it saturates near 14.3 and stays below the real-to-real baseline of 15.9 even at one twentieth of β^*. Over the same range the cross-visit covariance error grows from 0.59 to 0.69. Marginal and mechanistic fidelity are insensitive to β, so the longitudinal covariance structure bears the cost of lowering it. No inverse temperature yields a cohort that is both as private as the real patients are from one another and faithful to their joint structure.

Neither lever settles the question completely, because every scheme compared so far is a finite-time Markov chain. The target itself is available in closed form. Completing the square in Eq. (1) gives Eq. (3). For unit-normalized memories π_r is exactly the isotropic Gaussian mixture $\sum_k q_k \mathcal{N}(\mathbf{m}_k, \beta^{-1}\mathbf{I})$ with $q_k = r_k / \sum_j r_j$. It can therefore be sampled ancestrally, by drawing a component and adding isotropic noise. There is no chain, no initialization, no burn-in and no discretization error. We generated a protocol-matched exact cohort (N=100 at β^*, reusing the ladder's seeded magnitude draw so that only the direction sampler differs) and evaluated it on the identical harness (Table S16). Every metric falls inside the range spanned by the four rungs: median relative error 1.68%, cross-visit Frobenius error 0.60, mechanistic overlap 0.91, median novelty 0.50, and median DCR 13.87. One hundred independently seeded exact cohorts place these quantities in narrow intervals, with median relative error 1.40% (fifth to ninety-fifth percentile 0.97 to 1.91%) and median DCR 13.95 (13.61 to 14.39). The agreement does not rest on a favorable seed. Over the same 40 train/holdout splits used for the ladder endpoints, the membership-inference AUC under exact sampling is 0.98, marginally above the published scheme's 0.97. The memorization is therefore a property of π_r at K=23, where every stored patient is a component center. It is not a property of any approximation used to sample it.

Table S16: **Sampler-correctness comparison.** Each rung holds β^*, the memory, and the decoding fixed and varies only the direction sampler. Fidelity metrics (median relative error, cross-visit correlation Frobenius error, mechanistic cloud overlap, median novelty) and privacy metrics (median distance to the closest real record, and repeated-split membership-inference AUC where evaluated) are reported per rung. The real-to-real DCR baseline is 15.9. The membership-inference attack was run on the two endpoints of the ladder and on the analytic reference, over the same 40 train/holdout splits. *Exact* is not a rung. It draws ancestrally from Eq. (3) with no chain, and serves as the analytic reference for the four sampled rungs. Over 100 independently seeded exact cohorts the median relative error is 1.40% (fifth to ninety-fifth percentile 0.97 to 1.91%) and the median DCR is 13.95 (13.61 to 14.39).

Rung	Sampler	MRE (%)	Cross-visit Frob.	Mech. overlap	Novelty	DCR	MIA AUC
V0	anchor, endpoint	1.31	0.65	0.89	0.51	13.97	0.97
V1	sphere init	1.53	0.57	0.89	0.50	13.96	n/a
V2	pooled ULA	2.02	0.59	0.88	0.50	14.15	n/a
V3	pooled MALA	1.89	0.62	0.92	0.52	14.33	0.96
Exact	analytic ancestral	1.68	0.60	0.91	0.50	13.87	0.98

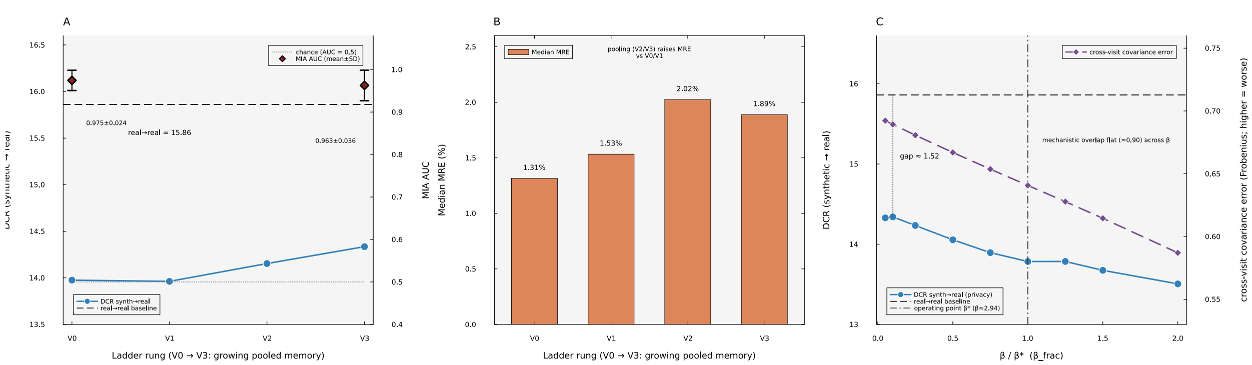

Figure S9: **Sampling correctness and the memorization/privacy relationship.** (A) Across the V0 to V3 ladder the distance to the closest real record (DCR, left axis) rises only slightly and stays below the real-to-real baseline, while the membership-inference AUC (right axis) remains near 0.96 to 0.97. (B) The fidelity cost of multi-chain pooling, as median relative error. (C) The inverse temperature swept across a 40-fold range around the operating point β^*. The DCR (privacy, left axis) rises as β falls but never reaches the real-to-real baseline. The cross-visit covariance error (right axis) grows, and the mechanistic overlap stays flat.

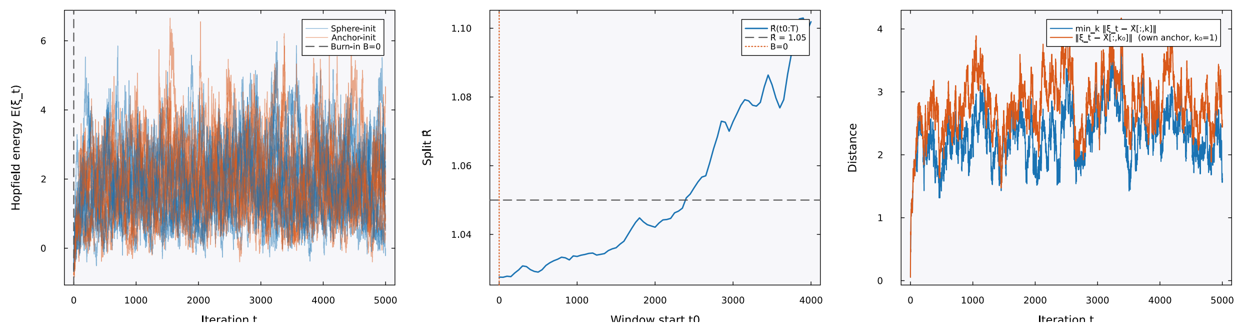

Figure S10: **Mixing diagnostic at β^*.** Energy traces, the between-chain $\hat{R}$ statistic, and the anchor-distance trajectory for sphere-initialized Langevin chains. The target distribution at β^* is multimodal, with one component centered on each stored pattern. Metropolis acceptance near 0.999 indicates negligible discretization bias.